

Enhancing medium voltage underground circuit design: Assessing limitations, thermal influence, and accurate modelling

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ABSTRACT

The research article conducted a study aimed at elucidating the limitations present in existing international standards related to the permissible quantity of medium voltage circuits that can be placed within a single trench. The study also explored important considerations for underground circuits in the context of renewable energy projects, particularly focusing on the crucial roles of capacity and load requirements in cable design. The research employed advanced finite element methods with Composition, Environmental, Structure, and Mechanism (CESM) and mathematical modeling to enhance existing procedures defined by international standards like IEC 60287, 60502, and 60228. The primary parameter of interest was the thermal influence of subterranean XLPE (cross-linked polyethylene) current capacity, considering various factors such as circuit quantities, distances, and depths. Notably, the study made several key findings, including a notable reduction in the current capacity of the underground circuits. This reduction in current capacity was found to vary, ranging from 100 % to 25.75 %. The reduction was attributed to factors such as native soil conditions, circuit separation. The findings furnish valuable insights into the challenges pertinent to medium voltage underground circuit design. Moreover, the paper establishes the accuracy of the proposed process, achieving a precision of 99.4850 ± 0.5150 % for 240 mm² and 99.3698 ± 0.6301 % for 630 mm².

1. Introduction

XLPE cables serve as a widely adopted insulation material for medium voltage applications, particularly within solar power plants for connecting conversion units, inverters, and wind farms. In wind farms, they facilitate interconnections from bus bars to medium voltage transformers in each wind turbine generator. Notably, XLPE cables have a standard maximum operating temperature of 90 °C. Over time, these cables experience degradation driven by four primary factors. Consequently, any cable system that carries a charge or current demonstrates a thermal response intrinsically linked to its progressive degradation. From this standpoint, four key characteristics underscore insulation degradation in XLPE cables; previous research reveals thermal degradation marked by oxidation induction periods, a result of prolonged exposure to elevated temperatures, including the highest operational temperature, overloading, or excessive ambient conditions [1]. This thermal aging could result in reduced mechanical strength and electrical properties, ultimately leading to the premature failure of the insulation.

For instance, the XLPE cables are extensively utilized for medium voltage insulation across various applications. However, the operational limitations and the subsequent degradation processes necessitate a comprehensive understanding of their behavior and properties.

XLPE insulation is susceptible to oxidation when exposed to oxygen in the environment, in that case, the combination of the oxygen inside the cable with the aluminum could react with the polymer chains, causing degradation of the insulation material over time with combustible gases as Hydrogen and hydrocarbons, besides, under anaerobic conditions, the results are CO and CO₂ [2]. Therefore, oxidation could expand the aluminum cable in the inside of the insulation, the result in the formation of free radicals, chain scission, and loss of mechanical and electrical properties degradation of partial discharge in the interior. The water Treeing as a phenomenon with the XLPE insulation and its degradation with the moisture influence in long terms. Therefore, the water content could lead to the formation of microscopic water-filled channels within the insulation. Electrical stress, with the magnetic and electric field gradients, it could accelerate the degradation of XLPE insulation with electrical stress in the insulation with the formation of

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Nomenclature			
<i>Symbol and acronyms list</i>			
$\Delta\theta$	Increase in temperature of the electrical conductor with respect to the ambient temperature (K)	k_p	Proximity factor
θ	Soil temperature under normal operating conditions ($^{\circ}\text{C}$)	k_1	Soil temperature characteristic
θ_c	Cable temperature ($^{\circ}\text{C}$)	k_2	Grouped circuits factors
φ_1	Ratio of losses in the metallic sheath with respect to the total losses in all the conductors in that cable. Zero (0) is considered because the cable is not armored	k_3	Depth of the circuits
φ_2	Ratio of armor losses and total losses in all cables. Zero (0) is considered because the cable is not armored	k_4	Thermal ground resistivity factor
ρ_T	Thermal resistivity of the insulation (m.K/W). It is 3.5 according to the IEC 60287 [4] for XLPE	L	Distance between the ground surface and the axis of the cable (mm)
B	Magnetic flux density	n	Number of circuits loaded
$\cos(\theta)$	Power factor	P	Transformer LV/MV active power
CESM	Composition, Environmental, Structure, and Mechanism	R	AC electrical resistance of the conductor at maximum operating temperature (Ω/m)
d_c	Cable diameter (mm)	R'	DC electrical resistance of the conductor at the maximum operating temperature (Ω/m)
D_e	External diameter of the cable (mm)	t_1	Insulation thickness (mm)
D	It is the fictitious diameter immediately below the jacket, in millimeters	T_1	Thermal resistivity between a conductor and its insulation (m.K/W)
D%	Condition of degradation of the underground cable (%)	T_2	Thermal resistivity between the insulation and its armor (m.K/W)
E	Electric field	t_3	Insulation of the cable jacket (mm)
F1	It is the depth from the ground level to the limit of the backfill	T_3	Thermal resistivity of the jacket (m.K/W)
F2	It is the distance from the backfill (first to the last ground layer)	T_4	Thermal resistivity between the surface of the cable and the medium (m.K/W) – three monopole cables in trefoil configuration
f	Electrical frequency (Hz)	V	Rated voltage
I	Rated current	W_d	Dielectric losses of the insulation (W/m)
k_s	Skin factor	x	Number of years in the age of the underground cable (years)
		XLPE	The single-pole cables used are made up of conductors insulated with polyethylene crosslinked

space charge accumulation, partial discharges, and the generation of localized heat. In this case, the water droplets move to the high electrical field, according to the Maxwell stress Continuous exposure to electrical stress can degrade the insulation material, leading to a decrease in its dielectric strength and eventual failure caused by the Triallyl Isocyanurate called TAIC [3]. On the other hand, we could change the evaluation of the mechanical stress; the mechanical damage on the insulation as the cracks, it allows the water content or moisture, reduced dielectric strength, and insulation failure. Finally, the external contaminants, such as dirt, chemicals, or oils, can have a detrimental effect on XLPE insulation; it could interact with chemical reactions or physical changes that degrade the insulation's properties.

For instance, to reduce this influence, two international standards improve the underground cable design: IEC 60287-1 and IEC 60287-2. The main characteristics are the following:

- The IEC 60287-1 provides guidance on the design of power cables, including underground cables with a comprehensive framework for medium voltage cable design, with a limitation according to the specific guidance on the design of cables for applications or in specific environments. For example, it does not provide specific guidance on the design of cables for use in highly corrosive, hazardous environments, or desert conditions with high values of soil with thermal resistance; besides, a limitation associated to the static operating environment. In this case, the IEC 60287-1 hasn't included the dynamic factors such as the effects of wind, vibration, or other environmental factors on cable performance [4].
- The IEC 60287-2 provides guidance on the calculation of thermal resistance for power cables, with a comprehensive framework for thermal resistance calculations. It has two limitations, it assumes a steady-state thermal environment, therefore, the dynamic factors are not considered, for instance the thermal cycling as thermal stress. In

addition, this standard considers homogeneous soil or backfill environment, and it does not account for variations in soil thermal properties or temperature profiles, available in desert conditions. On the other hand, it doesn't consider the joints and terminations on thermal resistance, the impact is not considered in the standards [5].

Therefore, the limits of these standards are the quantity of the circuits in the same trench. All the international standards consider only two circuits in the same trench. However, the limitation of the area and the environmental impacts require to increase the quantity, to reduce the impact of the new infrastructure. The key aspects for the future of the design should consider the aspects in the design: The capacity and Load Requirements for renewable energy projects, such as wind farms or solar power plants, typically require substantial capacity and load-bearing capabilities. Designing medium voltage underground cables for these applications involves accounting for the power generation capacity and the projected load demand to ensure optimal cable sizing and capacity planning [4]. On the environmental point of view, the factors, including temperature fluctuations, moisture, and potential exposure to chemicals or corrosive substances with the thermal resistance of the soil are crucial to ensure long-term cable performance and durability in these challenging conditions [6]. In the system integration perspective, the voltage stability levels within the desired limits is critical for efficient power transmission and distribution in order to consolidate the energy losses during transmission and distribution with the proper conductor sizing, and optimal insulation materials helps minimize these losses and maximize energy efficiency, however, to reduce the impact in fault conditions, such as short circuits, and should incorporate appropriate protective devices like circuit breakers or fuses, in the limitation of the construction, by selecting suitable cable types, burial depths, and installation methods to ensure reliability, accessibility, and ease of maintenance and finally, the cost optimization, by balancing the

selection of cable materials, insulation types, and cable configurations with the overall project budget helps achieve an optimal solution without compromising reliability and performance, with the international standards mentioned before. The Spectral Elements Method (SEM) is a versatile numerical technique with potential applications in enhancing the accuracy of underground cable calculations within a single trench. This method employs spectral discretization to represent physical properties and geometries. By refining the spectral elements, future advancements can be pursued to refine its efficacy and precision in modeling cable systems, to solve electromagnetic fields in scattering and radiation problems [18].

1.1. Motivation

Currently, the international standards used in the design of medium voltage underground cables installed directly in the soil has the limitation of two circuits per trench as top quantities. However, in the PV solar plants and wind farms, from the substation to the inverters and wind turbine generators should share the same trench for 3–9 km as a common requirement. For that reason, the international standards should consider the mathematical modelling with the finite element validation and the effect of the limitations with the currently design; by selecting suitable cable types, burial depths, and installation methods to ensure reliability, accessibility, and ease of maintenance and finally, the cost optimization, by balancing the selection of cable materials, insulation types, and underground circuit configurations.

This research article has the following sections: Section II: Introduces the methodology for the analysis of the underground cable design and limits of the international standards with the failure analysis of the edge associated to the medium voltage XLPE cable design, with the IEC

standard procedure with the evaluation of the electromagnetic field, thermal aspects. The Section III allows to validate with the finite element modelling, for the cable from one circuit to eight circuits with the results in the current capacity and electric field. Besides, the comparison of the electrical test and simulation with the laboratory test for the degradation of the XLPE cables. The Section IV considers the discussion of the results with the literature and new improvements for the future of the underground cables XLPE design. Finally, the information of the conclusions and recommendation for future works.

2. Methodologies

2.1. Finite element methodology with CESM composition

In the creation of the methodology for the evaluation of the root cause analysis of the XLPE associated to thermal constraints due to the distance between the circuits; the aging of cables with XLPE-type insulation is related to temperature. This type of the cables contains antioxidants, which protect the XLPE from oxidation during the manufacturing process (extrusion/crosslinking), but also during the cable's useful life. About the degradation ratio, the speed at which this antioxidant consumed directly dependent on the working temperature of the XLPE cable. At its maximum temperature, the consumption rate of the antioxidant is calculated during the manufacturing process, to provide a useful life for the cable of at least 30 years.

The methodology is based on the composition of the Composition, Environmental, Structure, and Mechanism CESM, proposed with Bunge [9], it created regarding the following three steps, in Fig. 1.

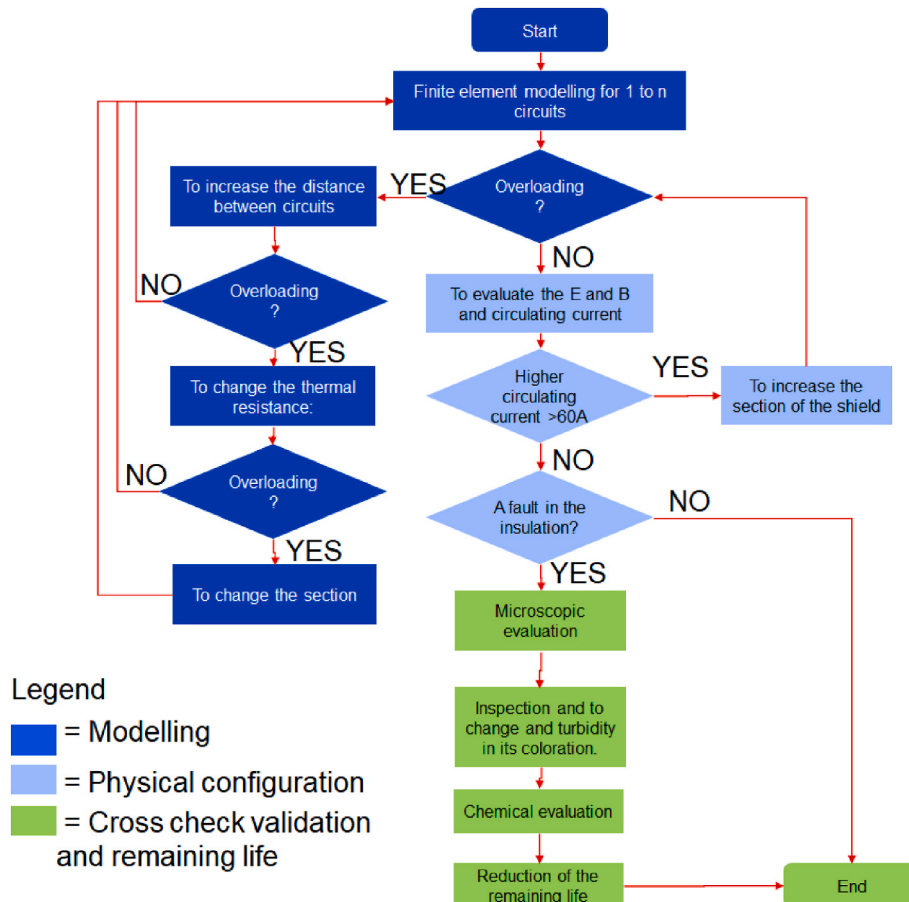


Fig. 1. Methodology for the evaluation of the thermal degradation analysis.

- The first step is based on the modelling (blue color in the Fig. 1): The finite element modelling for one to “n” circuits, the environment with the backfilled used in the circuit represented with the thermal resistance of the surface and the configuration of the circuits. The Composition used the information of the distance between the circuits, the thermal resistance of the backfill, and the section represented with the main characteristics of the cable as the outer sheath, aluminum sheath, buffer layer, insulation screen, insulation layer, conductor screen and conductor [10], in this case a validation of the overloading with the constraints of the maximum temperature in the underground cables (90°C); it should be used to increase the distance between the circuits or changing the thermal resistance of the backfill or changing the section of the underground cable.
- The second step is based on the physical configuration (sky blue color in the Fig. 1) based on the structure evaluation with the evaluation of the electric (E) and magnetic field (B) and circulating current in the shield, it reduces the current capacity, and it increases the temperature in the nominal current and conditions [7] with the temperature

125°C , 135°C , 150°C and 160°C and the evaluation with the Arrhenius graph.

- Finally, the cross-check validation (green color in the Fig. 1), it considers the mechanism as the main factor for the analysis of the failure mode with the evaluation of the effect over the XLPE components as the microscopic evaluation, inspections and turbidity in its coloration, chemical change due to thermal influence and the reduction of the remaining life.

2.2. Mathematical model for more than 2 circuits in the trench

In Fig. 2, our methodology significantly enhances the design process of medium voltage underground cables as per IEC 60287 [4,5] and IEC 60502 [12], particularly focusing on XLPE cables. Initially, we determine the maximum current demand based on connected loads, such as motors or transformers. In our case study, it evaluates a load linked to the medium voltage bus bar within a PV solar plant, connected through low voltage to medium voltage transformers.

The subsequent step involves calculating the medium voltage AC

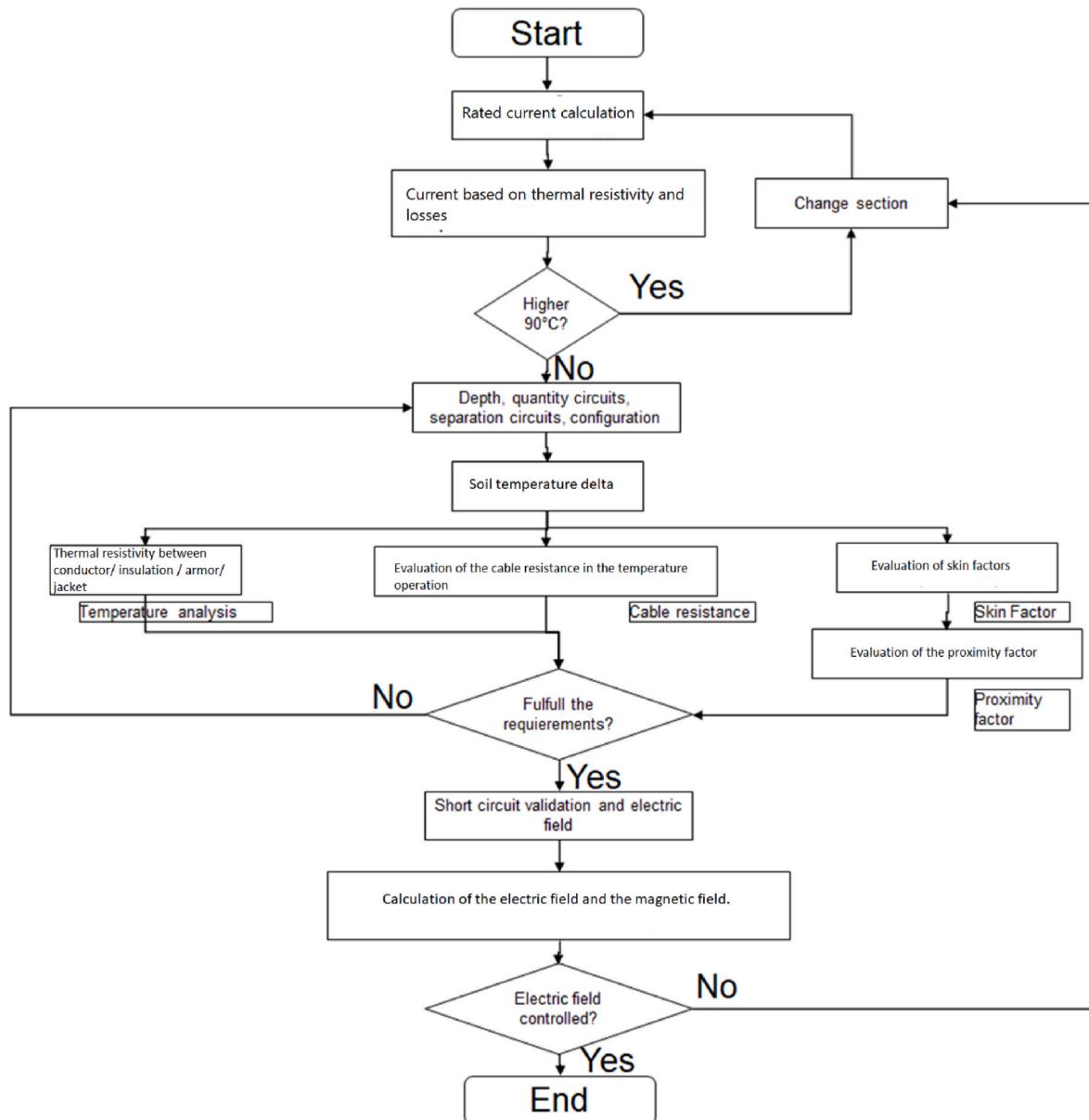


Fig. 2. Methodology for medium voltage underground cable design.

cables under dry-out conditions, considering the conductor's temperature concerning ambient conditions. This paper analyzes the number of circuits and their thermal resistivity, adhering to a maximum cable temperature of 90 °C. Our analysis encompasses factors like dielectric losses in insulation, losses in the metallic sheath relative to overall conductor losses, and armor losses compared to total cable losses.

Furthermore, we account for resistance, considering IEC 60287-1 [4], skin effect, proximity effect, and image001.png representing DC conductor resistance. We incorporate a constant mass temperature coefficient and the maximum operational temperature. Soil conditions necessitate correction factors applied in four stages: soil temperature, grouped circuits within the same trench, circuit depth, and thermal ground resistivity. All three phases, temperature analysis, cable resistance, and factors such as skin and proximity effects, are concurrently evaluated.

Subsequently, we assess fulfillment, including short-circuit validation and electric field lower than 5 kV/m, magnetic field strength H lower than 80 A/m and magnetic flux density lower than 100 μ T for the control within the cable and terminals [14], and the Peruvian recommendation limited 0.5 V/m for 1 μ T of magnetic flux. When selecting medium voltage cables for wind power plants, up to 60 kV, we rely on XLPE cables with current capacity guidance from IEC 60287, complemented by IEC TR 62095 technical reports. Dimensioning considerations encompass current capacity, configuration, installation dimensions, and cable materials.

Thermal resistance is a critical factor, guided by Section 3.2 of IEC 60287 3-1 and IEC 60287 2-1 tables. We select suitable backfill materials, in addition to native soil, to ensure efficient heat transfer. Various constraints, including maximum conductor current at 90 °C, insulation design, shielding, grounding systems, circulating current, contamination levels, humidity, and proximity to coastal environments, shape our design decisions.

Finally, we verify short-circuit capacity in accordance with IEC 60949 standards.

In this sense, the following are the criteria for the selection of the cables according to the actual IEC standard:

- Required current capacity (transmission scenarios), with the technical information on the installed cable (drawings and technical data sheet).
- Technical information on the installed terminals and the installed splices.
- Route map of the circuits, with types of trenches installed regarding to the detail plan of trenches of the circuits and the thermal resistivity of the soil.
- Maximum ambient temperature in the installation area.
- Information on significant heat sources that are close to the route of the cables.

In the renewable energy field, a prevalent practice involves the utilization of underground power lines. These subterranean lines are strategically positioned alongside roads, featuring aluminum XLPE (RHV) cables in sections of 95, 150, 240, 400, and 630 mm², designed for 20/34.5 (37.95) kV operation. They establish crucial connections from each wind turbine to the central Substation at 33 kV. Ensuring effective grounding, we employ an AWG 1/0 copper cable that interconnects the wind turbines with the Substation.

Moreover, within the same trench housing the underground power lines, we've established a robust communications network. This network relies on fiber optic cables to facilitate monitoring and control of the Wind Farm. The electrical grid operates at 33 kV and adheres to seven key electrical characteristics, detailed in Table 1:

- Rated voltage: 33 kV.
- Highest material voltage: 37.95 kV.
- Impulse test voltage: 170 kV.

Table 1

Characteristics of the cable used in the medium voltage system.

Characteristics	95 mm ²	150 mm ²	240 mm ²	400 mm ²	630 mm ²
Cable diameter (mm)	11.70	14.60	18.60	23.70	31.03
Exterior cable diameter (mm)	37.00	40.00	44.00	50.00	58.62
Weight (kg/km)	1.470	1.770	2.230	2.970	4.388
Resistance 60Hz at 90° (Ω/km)	0.410	0.264	0.160	0.0997	0.065
Reactance per phase (Ω/km)	0.210	0.199	0.186	0.174	0.165
Maximum current with three cables at 1 m depth to 25 °C	264	336	432	548	685

- Test voltage at 60 Hz: 70 kV rms.
- Short-circuit intensity of short duration (1s): 20 kA.
- Peak value of the short-circuit current: 40 kA.
- XLPE insulation shielded with copper tape in a helical configuration, unarmored, and with a robust 2 mm thick outer sheath colored in red. This sheath is crafted from a polyolefin mixture in accordance with UNESA 3305 C and UNE HD 620 standards, with aluminum cable sections ranging from 95 to 630 mm².

The calculation of the sections between wind turbines should consider the voltage drop, current density and optimization of losses and investment. It will be made with XLPE (RHV) 20/36 kV cable with sections 95, 150, 240, 400 and 630 mm² aluminum, adapting to the load in each section, so that the losses are less than 1.5 %, that the power carried by the cable does not exceed in any case 90 % of its capacity rated for installation conditions, and that the regulatory criteria for voltage drop (2.5 % maximum) are respected.

For the calculation of conductors, the maximum current of the LV/MV step-up transformer integrated in each PCS is considered in the pair of inverters for the 2 pairs, according to the rated current calculation in Eq. (1). The current is validated with the Table 2, from IEC 60502-2 [2], it has the current ampacity for the medium voltage conductors up to 400 mm², in laboratory condition [4].

$$I = \frac{S}{\sqrt{3} \times V \times \cos(\theta)} \quad (1)$$

P: Transformer LV/MV active power.

V: Rated voltage.

I: Rated current.

$\cos(\theta)$: Power factor

Therefore, it is necessary to apply the equation according to IEC 60287-1-1 in section 1.4.1.1 [3], regarding medium voltage AC cables in which dry-out condition, with the current based on thermal resistivity and losses in Eq. (2), [4].

$$I = \sqrt{\frac{\Delta\theta - W_d[0.5T_1 + n(T_2 + T_3 + T_4)]}{RT_1 + nR'(1 + \varphi_1 + \varphi_2)(T_3 + T_4)}} \quad (2)$$

Where,

$\Delta\theta$: Increase in temperature of the electrical conductor with respect to the ambient temperature (K)

n: Number of circuits loaded.

T_1 : Thermal resistivity between a conductor and its insulation (m.K/W).

T_2 : Thermal resistivity between the insulation and its armor (m.K/W).

T_3 : Thermal resistivity of the jacket (m.K/W).

T_4 : Thermal resistivity between the surface of the cable and the medium (m.K/W) – three monopole cables in trifoil configuration.

R: AC electrical resistance of the conductor at maximum operating temperature (Ω/m).

R': DC electrical resistance of the conductor at the maximum operating temperature (Ω/m).

Table 2

Current rating for single-core cables with XLPE insulation [2].

Section cable (mm ²)	Trefoil direct to ground	Flat spaced direct to ground	Trefoil in ducts	Flat touching ducts	Trefoil in air	Flat touching	Flat spaced
240	367	373	351	356	502	511	593
300	414	419	397	402	577	586	677
400	470	466	451	457	673	676	769

W_d : Dielectric losses of the insulation (W/m).

φ_1 : Ratio of losses in the metallic sheath with respect to the total losses in all the conductors in that cable. Zero (0) is considered because the cable is not armored.

φ_2 : Ratio of armor losses and total losses in all cables. Zero (0) is considered because the cable is not armored.

Later, each term of the Eq. (2) is calculated separately, with the soil temperature delta with the Eq. (3), associated to the difference between the maximum temperature of the conductor under conditions operating conditions with $\Delta\theta$, it is 90 °C according to Table 4, IEC 60502-2 [2], and the ground temperature. Besides, d_c is obtained from IEC 60228 [13], it is resulting in 17.8 mm and 29.3 mm for the 240mm² and 630mm² calibers respectively [5,6].

$$\Delta\theta = \theta_c - \theta \quad (3)$$

$$T_1 = \frac{\rho_T}{2\pi} \times \ln \left(1 + \frac{2t_1}{d_c} \right) \quad (4)$$

$$T_3 = \frac{\rho_T}{2\pi} \times \ln \left(1 + \frac{2t_3}{D_a} \right) \quad (5)$$

$$t_3 = 0.035D + 1.0 \quad (6)$$

$$T_4 = \frac{1.5 \times \rho_T}{2\pi} \times \left(\ln \left(\frac{2 \times L}{D_e} \right) - 0.63 \right) \quad (7)$$

$$u_{(section)} = \frac{2L}{D_e} \quad (8)$$

Where:

θ : Soil temperature under normal operating conditions (°C).

θ_c : Cable temperature (°C)

T_1 : It is calculated from the equation indicated in IEC 60287-2-1 [3], with respect to monopolar cables.

ρ_T : Thermal resistivity of the insulation (m.K/W). It is 3.5 according to the IEC 60287 [4] for XLPE.

t_1 : Insulation thickness (mm), it is 8.0 mm for the cable 18/30 (36 kV), according to the IEC 60502-2 [2].

d_c : Cable diameter (mm).

T_3 : It is calculated from IEC 60287-2-1, (m.K/W).

t_3 : Insulation of the cable jacket (mm).

D : It is the fictitious diameter immediately below the jacket, in millimeters (specified in Annex A of the IEC 60287-2-1) [5], being 17.5 mm (10mm²) and 28.3 mm (50mm²).

L : Distance between the ground surface and the axis of the cable (mm). L is considered 800 mm according to the installation conditions of IEC 60502-2 [12], it should be considered that for the project an installation depth of 900 mm is considered according to regional standards, instance for Colombia is the NTC2050.

D_e : External diameter of the cable (mm). It is obtained from the cable

Table 3

Relative permittivity and loss factors for the insulation of XLPE cables.

Type of cable	ϵ	$\tan \delta$
XLPE: Up to and include 18/30 (36) kV cables unfilled.	2.5	0.004
XLPE Greater than 18/30 (36) kV cables (unfilled)	2.5	0.001
XLPE: Greater than 18/30 (36) kV cables (filled)	3.0	0.005

Table 4

Comparison of the dielectric losses for the operation voltage of 30 kV and 33 kV.

Cable section	150	240	400	630
Relative permittivity in the insulation "e"	2.5	2.5	2.5	2.5
External diameter in the insulation "Di" (mm)	32,1	37,1	42,1	57,2
External diameter in the cable "dc" (mm)	16,1	21,1	26,1	29,8
Capacitance "C" (F/m)	3,62E-09	4,43E-09	5,23E-09	5,23E-09
Tanδ	0,004	0004	0,004	0004
W	376,99	376,99	376,99	376,99
Dielectric losses "Wd" (W/m) [30 kV]	1,6E-06	2,0E-06	2,4E-06	3,0E-06
Dielectric losses "Wd" (W/m) [33 kV]	2,0E-06	2,4E-06	2,9E-06	3,1E-06
Variation (%)	21 %	21 %	21 %	21 %

data previously calculated at T1 and T3 ($dc+2 \cdot t_1+2 \cdot t_3$), resulting in 37.0 mm and 49.3 mm respectively.

Besides, the resistance (R) is calculated from IEC 60287-1-1 in the Eq. (9) [4], with δ_s as the skin effect factor and the proximity effect factor δ_p ; and the R_0 is the DC resistance of the conductor at 20 °C in Ω/km , α_{20} is the constant mass temperature coefficient at 20 °C, and θ is the maximum operating temperature in °C [7].

$$R = R_0 \times (1 + \alpha_{20}(\theta - 20)) \times (1 + \delta_s + \delta_p) \quad (9)$$

$$\delta_s = \frac{x_s^4}{192 + 0.8 \times x_s^4} \quad (10)$$

$$x_s^2 = \frac{8 \times \pi \times f \times 10^{-7}}{R_0 \times (1 + \alpha_{20}(\theta - 20))} k_s \quad (11)$$

$$\delta_p = \frac{x_p^4}{192 + 0.8 \times x_p^4} \frac{d_c}{s} \left(0.312 \times \left(\frac{d_c}{s} \right)^2 + \frac{1.18}{\frac{x_p^4}{192 + 0.8 \times x_p^4} + 0.27} \right) \quad (12)$$

$$x_p^2 = \frac{8 \times \pi \times f \times 10^{-7}}{R_0 \times (1 + \alpha_{20}(\theta - 20))} k_p \quad (13)$$

Where:

f : Electrical frequency (Hz).

k_s : Skin factor, for round and stranded aluminum cable has the value of 1. It could be calculated according to the IEC 62281 with the Eq. (14) with the external diameter of cable without the shield (Di) and the external diameter of the cable (Dc).

k_p : Proximity factor, for the round and stranded cable in condition of dried and impregnated or not has the value of 0.8, [8].

$$k_s = \left(\frac{Dc - Di}{Dc + Di} \right) \left(\frac{Dc + 2Di}{Dc + Di} \right)^2 \quad (14)$$

Finally, in the Eq. (15), W_d factor is composed of the frequency, voltage reference to ground (U_o^2), $\tan \delta$ of 0.001 for XLPE cable 18/30 (36)kV, and the capacitance (C), according to the Eq. (16) with the relative permittivity (ϵ) of 2.5, the external diameter of cable without the shield (Di) and the external diameter of the cable (dc) [9].

$$W_d = 2 \times \pi \times f \times C \times U_o^2 \times \tan \delta \quad (15)$$

$$C = \frac{\varepsilon}{18 \ln\left(\frac{D_i}{D_c}\right)} \times 10^{-9} \quad (16)$$

For instance, the Fig. 3 shows the description for the evaluation of the underground cables. Regarding to the conditions of the soil, the correction factors should be applied in four stages, with the average thermal soil calculated according to the Eq. (17) [10],:

- Soil temperature characteristics for k1.
- Grouped circuits in the same trench in k2.
- Depth of the circuits in the same trench in k3.
- Thermal ground resistivity in the soil in k4.
- F1: It is the depth from the ground level to the limit of the backfill, according to Fig. 3.
- F2: It is the distance from the backfill (first to the last ground layer), according to Fig. 3.

$$R_p = \frac{R_1 \times F1 + R_1 \times F2}{F1 + F2} \quad (17)$$

Finally, the validation for the induced voltage for the underground cables with the connection in just one point with surge arrester considered in the opposite side has the Eq. (18), Eq. (19) and Eq. (20), with ω is the angular frequency of the system, S is the distance between the cable axis, d is the geometric diameter of the shield and the current per phase I_a , I_b and I_c [11].

$$E_a = j\omega l a (2 \times 10^{-7}) \left(\left(-\frac{1}{2} \right) + \left(\frac{j\sqrt{3}}{2} \right) \right) \log(2S/d) \quad (18)$$

$$E_b = j\omega l b (2 \times 10^{-7}) \log(2S / d) \quad (19)$$

$$E_c = j\omega l c (2 \times 10^{-7}) \left(\left(-\frac{1}{2} \right) + \left(\frac{j\sqrt{3}}{2} \right) \right) \log(2S/d) \quad (20)$$

Therefore, it validates the collector underground lines, with the different magnitudes exposed by columns, more than meet the indicated calculation criteria, plus the maximum losses due to "Joule effect".

About the constraints, the results should consider the following:

- The maximum admissible permanent current $< I_{\text{max}}$ admissible by the cable.
- The voltage drops: must be less than 2.5 %.
- The power losses due to “Joule effect” $< 3\%$.
- The maximum short-circuits current supported by the conductor $>$ Calculated short-circuit current.

Calculations justifying the section chosen in each of the sections of the circuits of the projected medium voltage underground network, therefore, the two new parameters have defined in the [Table 2](#).

About the dielectric losses, according to the IEC 60287-1-1 [4] the

equation associated with the losses is indicated in the Eq. (21), with the Capacitance (C) in the Eq. (22).

$$W_d = w * C * U_o^2 * \tan \delta \quad (21)$$

$$C = \frac{\varepsilon}{\ln \frac{D_i}{d_c}} * 10^{-9} \quad (22)$$

Where: w : It is $w = 2 \times \pi \times f$.

C: It is the capacitance per unit (F/m).

U_o^2 : It is the voltage per phase square.

$\tan \delta$: It is the losses factor in the insulation according to Table 3.

ϵ : It is the relative permittivity according to the insulation in Table 3.

D_i : It is the external insulation diameter without screen (mm).

d_c : It is the external insulation diameter with the screen (mm).

Besides, with the current capacity of underground lines for medium voltage, the electric field, the circulating current, and dielectric losses are increasing in the insulation caused by the dipolar rotation. It is an interaction of the molecules during the movement and in the alignment with the oscillation of the electric field, with the rotational movement to obtain the trajectory of the field, in this case the friction grows, by the energy transfer between them, in the Eq. (22). It means more electric stress near to the cable and lower in the external diameter of the cable.

If the electric stress is continuous, then the cable insulation has degradation in long terms with the Eq. (23). In this case, the electric field produces electric loads in the middle of the insulation, changing the dielectric parameters of the material and produces irreversible damage.

$$E_r = \frac{dV}{dr} = \frac{V}{r \times \ln \left(\frac{R_0}{R_i} \right)} \quad (23)$$

Where:

E_r It is the electric stress with a radio "r" (kV/mm).

V: It is the voltage applied (kV).

r: It is half of the diameter associated to the electric field and the applied voltage (mm).

R_i : It is the internal radius of the insulation (mm), with difference between sections according to [Table 4](#).

R_0 : It is the external radio with the insulation (mm), with difference between sections according to Table 4.

Besides, according to the voltage applied, and elevated temperature of the cable, produce an early degradation of the cable due to special load distribution. It means a storage between the interface crystalline-amorphous insulation material, in the micro structural level.

This energy transfer reduces the life cycle of the insulation according to the Inverse power law according to the V-t test, in this case, the “V” is the electrical stress (kV/mm) (Eq. (24)), over the insulation in “n” (life cycle exponent) thought the time of the stress in “L” hours, according to the constant “k” associated to the XLPE cable [6], in the Eq. (25) and Eq. (26).

$$V^n * t=k \tag{24}$$

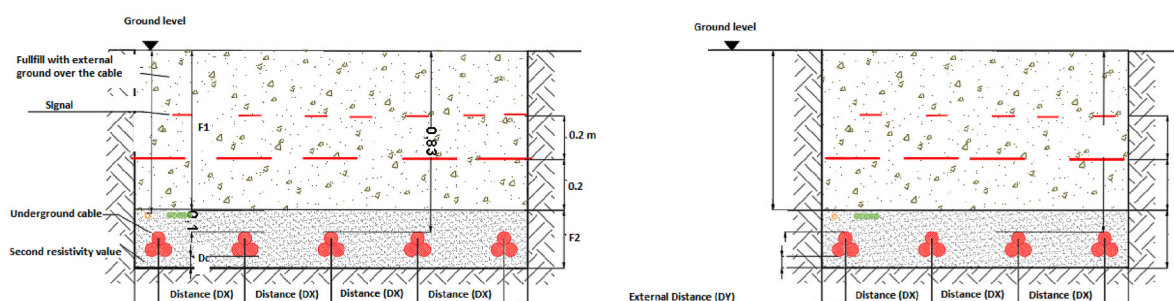


Fig. 3. Description of the F1 and F2 distance in the trench for the evaluation.

$$V_1^n * t_1 = V_2^n * t_2 \rightarrow n = \frac{\log \frac{t_2}{t_1}}{\log \frac{V_1}{V_2}} \quad (25)$$

$$L(V) = kV^{-n} \quad (26)$$

2.3. Degradation analysis with arrhenius equation

The results of the Boltzmann distribution law and the activation of the energy for thermal processing represented by the Eq. (27) and Eq. (28). It has considered the influence of the RT kinetic energy factor; due to the temperature influences with low Kinetic energy ratio with higher rate value with an accelerated reaction of thermal degradation [7].

$$k = A \times e^{-\frac{Ea}{RT}} \quad (27)$$

$$\ln(k) = -\frac{Ea}{RT} \times \ln(A) \quad (28)$$

Where:

k: It is the rated constant. It has the influence of the temperature. Usually, the authors [8], [10], recommend the representation $\ln(k)$ vs $1/T$, as a reference of the Arrhenius plot [8].

Ea: It is the energy activation.

R: It is the gas constant.

T: It is the Kelvin temperature of the process.

A: it is the Arrhenius factor.

3. Case study for underground XLPE thermal degradation

The case study has the installation of the cables from one circuit in the trench to eight circuits according to Fig. 4. A). Besides, the finite element model considered for the evaluation of the XLPE cable according to the following components in Fig. 4 B), as cable, shield, insulation, insulation screen, concentric wire, jacket, and overall diameter.

3.1. Finite element modelling

The effect of the grouping of the collector circuits in the trenches was analyzed, in order to show the degree of decrease in the transmission capacity of the power cables as the number of circuits installed in the same trench increases. The transmission capacity of collector circuits depends on the thermal resistivity of the ground, depth, cable installation spacing, and circuit grouping. Besides, the transmission capacity is a thermal phenomenon, its calculation conducted by means of a thermal balance in accordance with the IEC standard 60287. Therefore, heat sources close to power cables could be considered, the more heat sources

close to the power cable, the transmission capacity decreases. According to the technical specification of the power cable [4], the cable RHE-OL XLPE 630 mm² installed directly buried at a depth of 1.0 m, in a soil with uniform thermal resistivity of 1.5 K.m/W at 25 °C, has a transmission capacity of 564 A. Under the same conditions, in Fig. 20 shows the Cymcap simulation of the same power cable, obtaining a transmission capacity of 563 A; with the Fig. 5 A), therefore, the Cymcap calculation validates the manufacturer's information for a power cable. In this case, this model takes advantages of the results of the electric field, the numbers of cables and the distance and the insulation properties for XLPE, all covered conductors have an electric field reduction ratio of 0,983, but when one conductor is exposed, this ratio drops to 0, 9716 [15].

To evaluate the effect of the grouping of collector circuits, the most critical point of heat exchange was considered, which corresponds to the simulation point (8 grouped circuits), for which the thermal resistivity values of the ground obtained in test pit, proceeding with the simulation of a single circuit to which the other circuits were progressively added until obtaining the critical grouping of 8 circuits. As the first step, a circuit with RHE-OL XLPE 630 mm² cable, a transmission capacity of 350 A obtained, as shown in Fig. 5. B).

In the previous figure, due to the characteristics of the terrain (pit C-WII-04) and the ambient temperature. Therefore, the current capacity was reduced to 350 A, which represents 62.2 % of the capacity under normal conditions. standard with a uniform thermal resistivity of 1.5 K. m/W and with a ground temperature of 25 °C, which shows that increasing the thermal resistivity from 1.5 K.m/W to 4.892 K.m/W significantly reduces the cables' ampacity. Then, if we add a circuit with the same cable (RHE-OL XLPE 630 mm²) installed 250 mm from the previous circuit, the transmission capacity reduced to 268 A, which represents 76.6 % of the capacity of a circuit. when it is the only one installed in the trench, in Fig. 6.

In the evaluation of three circuits with the same cable (RHE-OL XLPE 630 mm²) separated by 250 mm, it is observed that the transmission capacity of the central circuit (circuit 2 of Fig. 23) is reduced to 187 A, which represents a 53.04 % of the capacity of a circuit when it is the only one installed in the trench, due to the presence of the lateral circuits, while the adjacent circuits are limited to a capacity of 252 and 250 A, respectively, in the Fig. 7. Considering all eight circuits of the simulation, the transmission capacity of the central circuits the capacity of the middle point reduced to 145 A, which represents 41.4 % of the capacity of a circuit when it is the only one installed in the trench, while the lateral circuits limit its transmission capacity.

Besides, in Fig. 8., as described above, as more circuits are grouped in the trench, the transmission capacity of the cables is increasingly reduced, even to only transport 41.4 % compared to if the circuit were the only one installed in the trench, it could cause them to be unable to

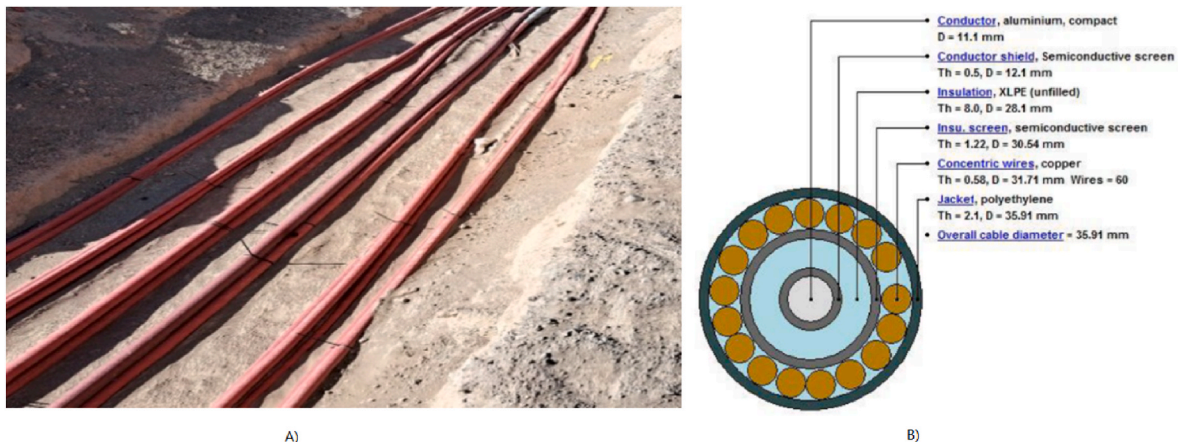


Fig. 4. A) Configuration, B) Composition of the XLPE cable, dimensions, and characteristics.

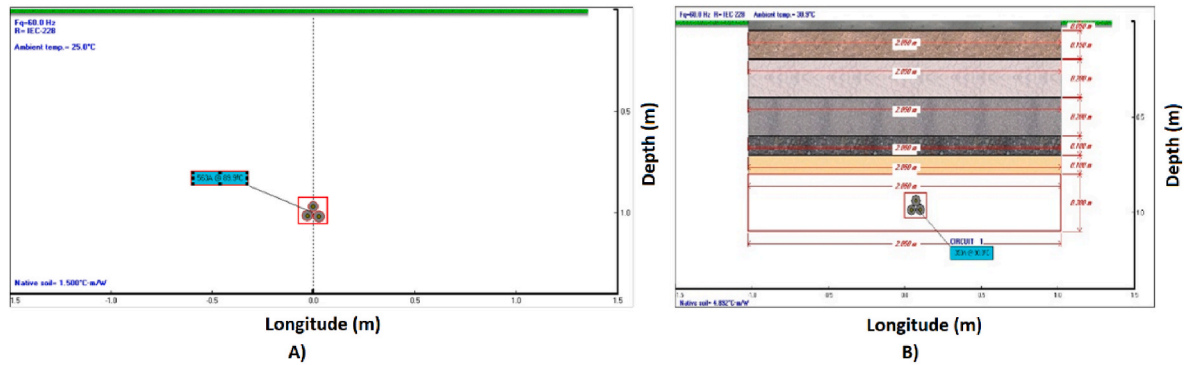


Fig. 5. A) Current Capacity of the XLPE with one type of soil: 563A with external soil, B) The evaluation of the common engineering associated to the one circuit XLPE with more than two soil layers, with a current capacity of 350A with native soil.

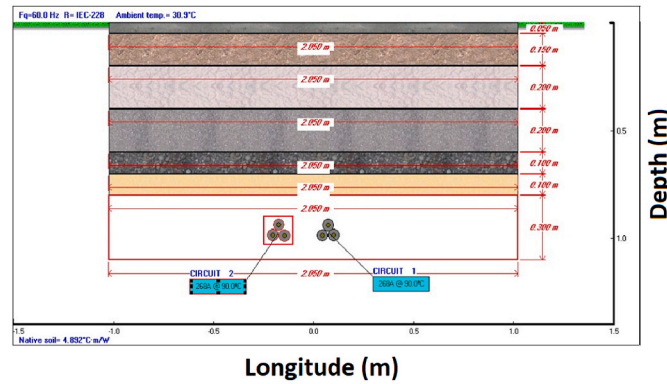


Fig. 6. Two circuit modelling separated 15 cm between circuits.

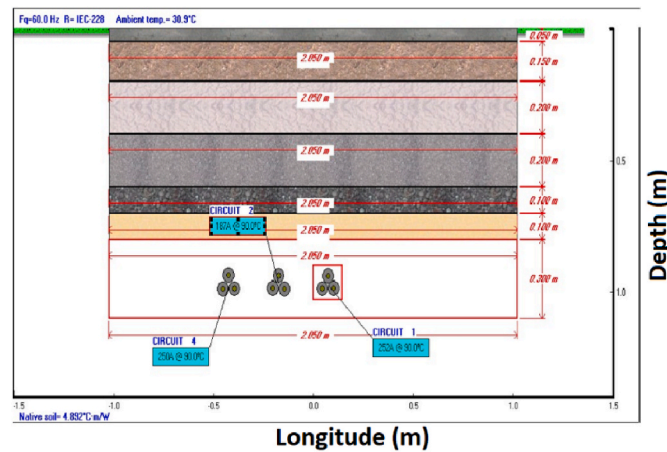


Fig. 7. Three circuit modelling separated 15 cm between circuits.

meet the capacity required by the system without exceeding its maximum operating temperature of 90 °C.

3.2. Failure analysis for the underground cable, with the distance of 15 cm per circuit recommended with IEC international standards

For this subsection, a specimen extracted in immediate post-fault times was available, which contains both the epicenter of the fault zone, as well as its adjacencies. It cuts in all directions, methodical quartering, and sectioning into slices $\approx \leq 1.0$ mm thick, randomly extracted. In Fig. 9 A), the radiographic indicates a venting channel skewed to one side. It observed that the epicenter of the fault is located

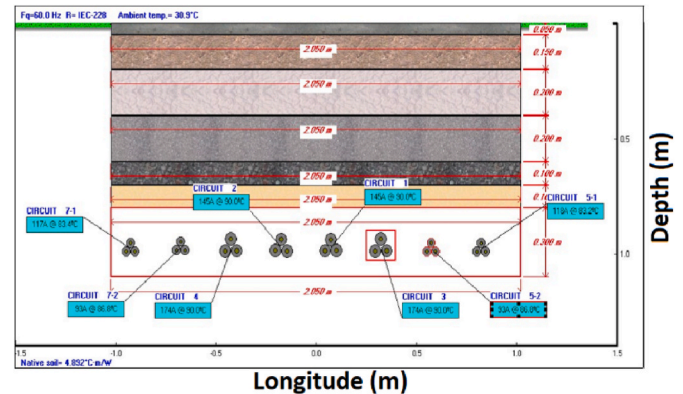


Fig. 8. Eight circuit modelling separated 15 cm between circuits: 4 Circuits with 630mm² in the middle and four circuits of 240mm² in the right and left position.

exactly on the helical projection of the copper tape of the screen (currently volatilized), arranged for the retention of the filaments, and an increase in the conductive section in Fig. 9B).

In the analysis, a clean perforation is described in Fig. 10, without the presence of an entry of any foreign material, therefore, the perforation shows an outward fold as a product of an internal accumulation of gases. A microscopic inspection shows the physiognomy of the perforation on the sample cover.

Exploding the sample, it could be seen that indeed, as in all the previous samples, the copper tape was located exactly above the epicenter of the fault, and that it, because of temperature and the consequent Radial expansion of the XLPE has undermined the inner side of the outer shell, confirming the influence of the copper tape in the fault sector. In Fig. 10, three marks on the external semiconductor, which copy the expansion suffered by the filaments of the screen, when dragged by the violence of the failure event during the release of gases. Continuing with the analysis of the failure, the semiconductor layer was removed, it is observed that the release of energy during the failure event found low intensity, due to the small radius of action of the carbonization on the insulation surface. primary. For instance, the Nanojets are linked to the subwavelength light focusing feature; tuning the parameters results in slowly diffracting beam propagation and high field confinement [19]; therefore, by making 0.20 ± 0.05 mm transversal cuts on the insulation, it could evaluate no defects in its construction. These defects correspond to protrusions or sinkholes of the insulation on the area of the external semiconductor layer. Despite its transparency, the characteristic yellowish color of an overheating of the XLPE observed, constituting this factor, fundamental evidence in any search for the root cause of failure.

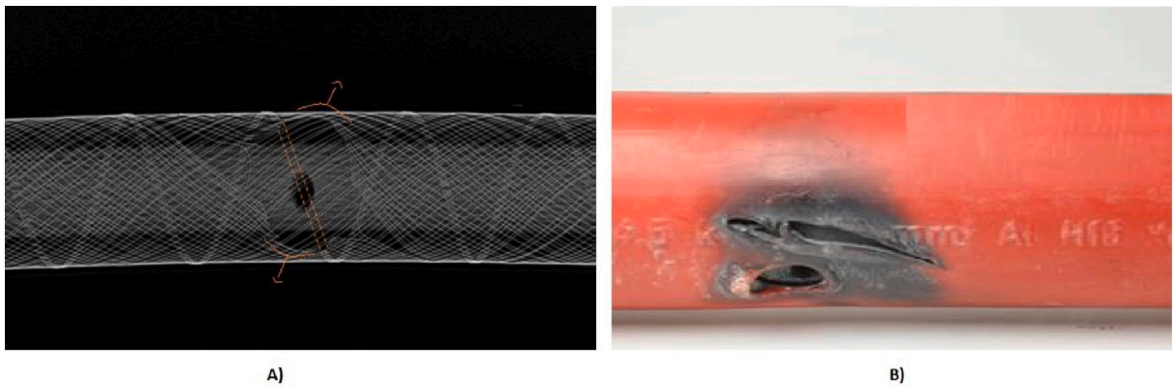


Fig. 9. A) Radiography of the failure. B) Real case study fault.

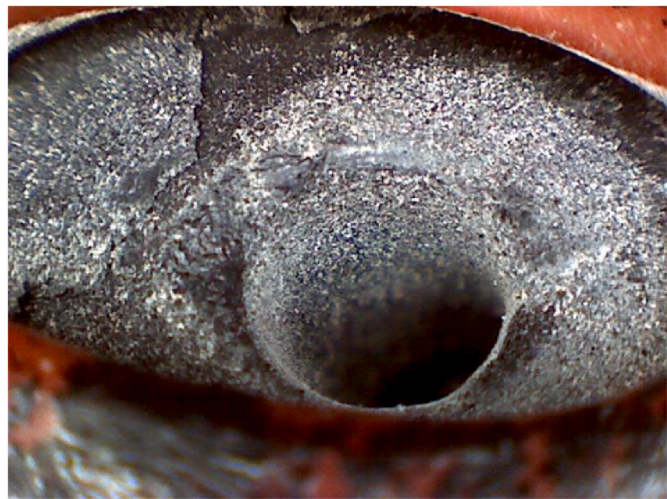


Fig. 10. Microscopic evaluation with physiognomy.



Fig. 11. Change and turbidity in its coloration.

In the evaluation of the profiles, the measurements of the thickness of the insulation and semiconductor layers, both internal and external. Therefore, it didn't show significant deformations, and no objections could raise regarding the constructive quality of the cable. In the Table 5, the average of the values is 8.005 ± 0.001 mm, the 50 % percentile is 7.991, 90 % percentile is 8.187, 10 % percentile is 7.884, the minimum value is 7.882 and the maximum value is 8.199, in the Table 4.

After of the methodical sectioning, and selective opening of five different sections of the sample, and more than fifty partial cuts subjected to microscopy/X-rays, the consequences of an evident degradation of the test tubes due to exposure are clearly observed on the XLPE insulation. prolonged thermal during its stage in service, characterized by the change and turbidity in its coloration, in Fig. 11.

About the partial discharge test, it found voltage levels from 1.0Uo to 1.7Uo; besides, the partial discharge activity is witnessed by increasing the test voltage up to 1.7Uo, increasing the resolution of the resulting phase pattern in normal fault condition, in lightning condition the value

of 15.571 % is the likelihood of a stroke current less than 12.52 kA. Furthermore, there is a 1 in 0.0033 % chance that there will be one failure annually. To reduce potentially harmful overvoltage, an earth-termination mechanism under high lightning currents was investigated. Many writers carried out this theoretically in the frequency and temporal domains. To find the soil resistivity layer in an earth-termination system, the earth electrode's length should be computed based on the LPS class. The electrode length is 50 m and the resistance of 2000 Ω m for class I [16], at least. However, for the evaluation of partial discharges we will base ourselves on the highest test voltage level 1.7Uo, where the resolved phase scheme is representative of partial discharges, showing partial discharges with a trend of continuously developing void failure, it could locate in the internal semiconductor of the MV cable. As a summary, the partial discharge is 200 with a maximum value of 3800 pC, in the Fig. 12 A). In the Fig. 12 B) is associated to the reflectometry in the MV cable, the detection of the fault allows to provide the areas with the highest incidence of partial discharges. In the reflectometry it observed that this fault is shown in an incipient way at the beginning of the cable, showing a normal behavior, since this is shown more frequently in the terminals.

Table 5
Insulation evaluation per quadrants.

Insulation sample	1st Quadrant	2nd Quadrant	3rd Quadrant	4th Quadrant
1	7.929	8.002	8.012	8.182
2	8.191	7.822	8.199	7.891
3	7.921	7.921	7.927	7.991
4	8.122	7.877	7.991	8.099

3.3. Results of the mathematical model with the finite element results

For the calculation of underground cables, the maximum current of the LV/MV step-up transformer integrated in each transformer, it considers a power per pair of inverters (each cabin has 2 pairs of inverters) of 4,240kVA, therefore, for each unit, it has a total of 8,480kVA at 25 °C.

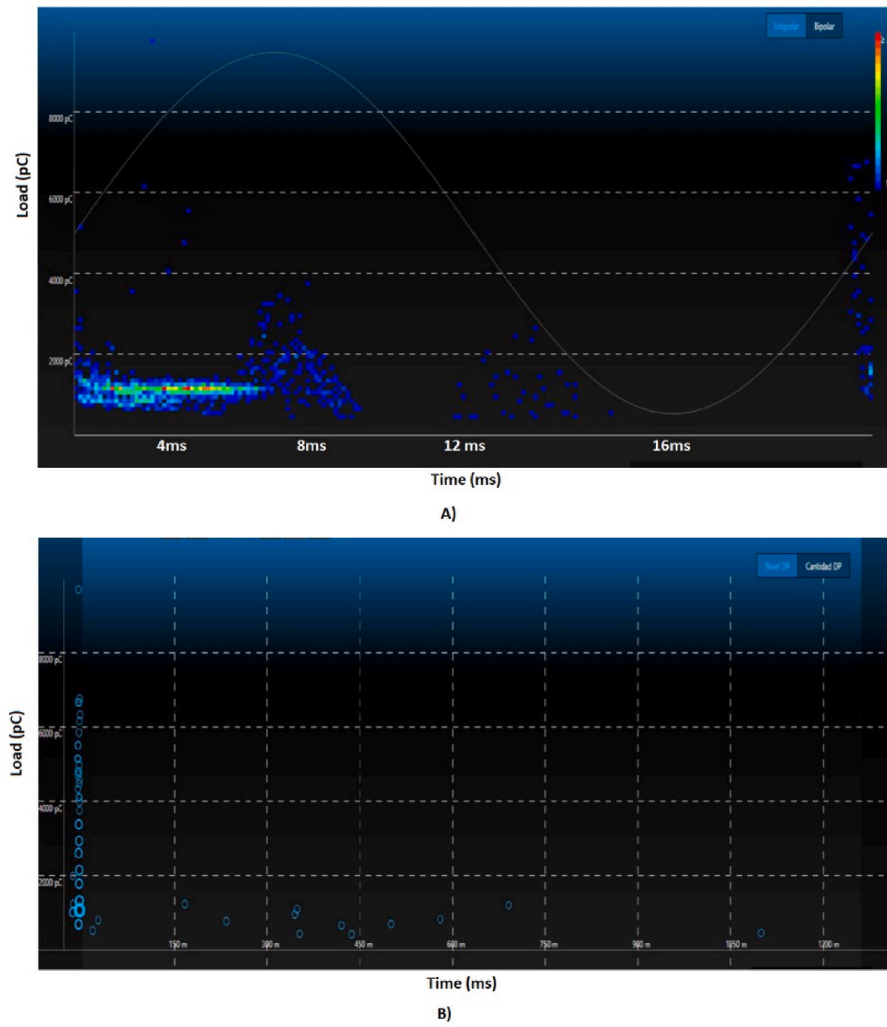


Fig. 12. A) Partial discharge of 1.7Uo at 34 kV. B) Location of the partial discharge in the underground cable.

therefore, it obtains 8.056 kW with a power factor 0.95 in Eq. (29), with the soil temperature delta in the Eq. (30), with the thermal resistivity for 240 mm² in Eq. (31) and 630 mm² in Eq. (32).

$$I = \frac{8.056 \text{ kW}}{\sqrt{3} \times 33 \text{ kV} \times 0.95} = 148.36 \text{ A} \quad (29)$$

$$\Delta\theta = 90 - 20 = 70^\circ \text{ C} \quad (30)$$

$$T_{240} = \frac{3.5}{2\pi} \times \ln\left(1 + \frac{2 \times 8}{17.8}\right) = 0.357 \text{ m.K} / \text{W} \quad (31)$$

$$T_{630} = \frac{3.5}{2\pi} \times \ln\left(1 + \frac{2 \times 8}{29.3}\right) = 0.243 \text{ m.K} / \text{W} \quad (32)$$

For the case study, and according to IEC 60502-2, the characteristics are the following: Soil temperature 20 °C, burial depth of 800 mm and thermal resistivity of 1.5 K m/W, with to verify the ampacity results of the 240 mm² and 630 mm², according to the Table 6.

In this case, the thicknesses of the jacket for the calibers evaluated in the Eq. (33) and Eq. (34), for 240 mm² and 630 mm².

$$t_{3(240 \text{ mm}^2)} = 0.035D + 1 = 0.035 \times 17.5 + 1.0 = 1.6 \text{ mm} \quad (33)$$

$$t_{3(630 \text{ mm}^2)} = 0.035D + 1 = 0.035 \times 28.3 + 1.0 = 2.0 \text{ mm} \quad (34)$$

Finally, the parameter D'a is obtained from the cable data in T1 (dc + 2 · t1), it is resulting in 33.8 mm for 240 mm² Eqs. (35), and 45.3 mm

Table 6
Description of the cable for the analysis.

Description	240 mm ²	630 mm ²
Current (A)	347	571
Voltage (kV)	35	35
Temperature (°C)	90	90
Cable	Compact circular conductor of aluminum wires (Class 2)	
Shield	Aluminum	Aluminum
Insulation	XLPE	XLPE
Other	UV and humidity resistance	

for the 630 mm² section in Eq. (36) and temperature jacker in Eq. (37) and Eq. (38) respectively.

In this case, the thermal resistivity T₃ is equal to Eq. (35)

$$T_{3(240 \text{ mm}^2)} = \frac{3.5}{2\pi} \times \ln\left(1 + \frac{2 \times 1.7}{17.8}\right) = 0.081 \text{ m.K} / \text{W} \quad (35)$$

$$T_{3(630 \text{ mm}^2)} = \frac{3.5}{2\pi} \times \ln\left(1 + \frac{2 \times 19}{29.3}\right) = 0.075 \text{ m.K} / \text{W} \quad (36)$$

$$u_{(240 \text{ mm}^2)} = 2 \times \frac{800}{37} = 43.2 \quad (37)$$

$$u_{(360 \text{ mm}^2)} = 2 \times \frac{800}{49.3} = 32.5 \quad (38)$$

$$T_{4(240mm^2)} = \frac{1.5 \times 1.5}{2\pi} \times (\ln(2 \times 43.2) - 0.63) = 2.743 \text{ m.K/W} \quad (39)$$

$$T_{4(630mm^2)} = \frac{1.5 \times 1.5}{2\pi} \times (\ln(2 \times 32.5) - 0.63) = 2.538 \text{ m.K/W} \quad (40)$$

In the Eq. (39) and Eq. (40), when calculating T4 in this way, T3 should be multiplied by a factor of 1.6 according to IEC 60287-2-1. And the resistance according to the Eq. (9), R_0 is obtained from IEC 60228, therefore, it is equal to 0.125 Ω/km and 0.0469 Ω/km for 240mm² and 630mm² gauges, respectively. Besides, the constant mass temperature coefficient at 20 °C for the aluminum cable is 4.03×10^{-3} , and the maximum temperature of the conductor is 90 °C for XLPE cable, with the resistance of the cable in the Eq. (41) for 240mm², and 630mm² in Eq. (42), besides, the capacitance calculated in the Eq. (43) with Eq. (45) and Eq. (44) with Eq. (46) for 240 mm² and 630 mm² respectively.

$$R_{(240mm^2)} = 0.125 \times (1 + 4.03 \times 10^{-3}(90 - 20)) \times (1 + \delta_s + \delta_p) = 0.16 \times 10^{-3} \Omega / m \quad (41)$$

$$R_{(630mm^2)} = 0.047 \times (1 + 4.03 \times 10^{-3}(90 - 20)) \times (1 + \delta_s + \delta_p) = 0.064 \times 10^{-3} \Omega / m \quad (42)$$

$$C_{(240mm^2)} = \frac{2.5}{18 \ln\left(\frac{33.8}{17.8}\right)} \times 10^{-9} = 2.165 \times 10^{-10} \quad (43)$$

$$C_{(630mm^2)} = \frac{2.5}{18 \ln\left(\frac{45.38}{29.3}\right)} \times 10^{-9} = 3.18 \times 10^{-10} \quad (44)$$

$$W_{d(240mm^2)} = 2 \times \pi \times 60 \times 2.166 \times 10^{-10} \times 19.053^2 \times 0.001 = 0.0324 \quad (45)$$

$$W_{d(630mm^2)} = 2 \times \pi \times 60 \times 3.187 \times 10^{-10} \times 19.053^2 \times 0.001 = 0.0436 \quad (46)$$

Finally, the evaluation of the underground cable XLPE in the Eq. (2) with all the parameters allows to calculate the maximum current for the cables (3 cables grouped in 1 circuit in trefoil configuration), directly in the ground.

$$I_{(240mm^2)} = \sqrt{\frac{70 - 0.0324[0.5 \times 0.357 + 1(0 + 0.081 + 2.743)]}{0.161 \times 10^{-3}(0.357 + 0 + 0.081 + 2.743)}} = 368.9 \text{ A} \quad (47)$$

$$I_{(630mm^2)} = \sqrt{\frac{70 - 0.044[0.5 \times 0.243 + 1(0 + 0.075 + 2.538)]}{0.64 \times 10^{-3}(0.243 + 0 + 0.075 + 2.538)}} = 618.9 \text{ A} \quad (48)$$

Therefore, the Eq. (47) has the same value as the IEC 60502 for the XLPE insulation with aluminum core (367A) with an accuracy of 99.4850 ± 0.5150 % in 240 mm², compared with the Eq. (49). The error is provided with the recommendation of the international standard.

With the evaluation of the cable of 630 mm², the value of the aluminum core is 615 A, with the accuracy of the calculation is 99.3698 ± 0.6301 % in the Eq. (48) compared with the Eq. (50). It is calculated for the ground thermal resistance of 1.5 Km/W and the soil temperature of 31 °C, with the factor of the Table 7.

$$I_{(240mm^2)} = 368.9 \times 0.9400 = 346.8 \text{ A} \quad (49)$$

$$I_{(630mm^2)} = 618.9 \times 0.9226 = 571.0 \text{ A} \quad (50)$$

Furthermore, the factor corrections should be selected according to the location and technical characteristics, as follows:

Table 7

Reduction of the cable for the current capacity.

Section of the underground cable	Factors
120	0.9600
150	0.9550
185	0.9500
240	0.9400
300	0.9356
400	0.9313
630	0.9226

- K1: For the soil temperature, for 31 °C and the project is selected to 0.918.
- K2: For the circuit separation, the literature considers the Table 4. To evaluate the case study, we selected 5 and 6 circuit separation, with 0.4 m: K2 = 0.72 for 5 circuits and k2 = 0.70 for 6 circuits, Table 8.
- K3: For the deep of the circuit, according to the Table 9. It should be 1 m, and the factor is 0.97.
- K4: Thermal ground resistivity factor. It is according to the evaluation on field, with a value of 1.5 Km/W and the ground over the underground cable has a thermal resistivity of 1.65 Km/W, in order to calculate the average thermal resistivity of the soil, with the information of the Eq. (51), and the Table 5 for the selection of the factor according to K4 equal to 0.88, Table 10.

$$R_p = \frac{1.5 \times 0.74 + 1.65 \times 0.27}{0.74 + 0.27} = \frac{1.54 \text{ Km}}{W} \quad (51)$$

The current capacity of the cable according to the factors k1, k2, k3, k4 with the current capacity in the Eq. (52) and Eq. (53), with the results in the Eq. (54) and Eq. (55):

$$I_{(240mm^2)} = 346.8 \text{ A} \times 0.92 \times 0.75 \times 0.97 \times 0.88 = 203.93 \quad (52)$$

$$I_{(630mm^2)} = 571.0 \times 0.92 \times 0.70 \times 0.97 \times 0.88 = 313.21 \text{ A} \quad (53)$$

The induced voltage has the following results with the Eq. (54), Eq. (56), Eq. (57) over the results in the Eq. (58) and Eq. (59), lower to the electric field limited 0.5 V/m [14] for 1 μT of magnetic flux.

$$E_{a(240mm^2)} = E_{c(240mm^2)} = E_x \quad (54)$$

$$E_x = j2\pi 60 \times 148.36 \times (2 \times 10^{-7}) \left(\left(-\frac{1}{2} \right) + \left(\frac{j\sqrt{3}}{2} \right) \right) \log \left(2 \times \frac{49.2}{44.4} \right) = 0.009 \text{ V/m} \quad (55)$$

$$E_{b(240mm^2)} = j2\pi 60 \times 148.36 \times (2 \times 10^{-7}) \log \left(2 \times \frac{61.5}{44.4} \right) = 0.004 \text{ V/m} \quad (56)$$

$$E_{a(630mm^2)} = E_{c(630mm^2)} = E_y \quad (57)$$

$$E_y = 0.018 \text{ V/m} \quad (58)$$

$$E_{b(630mm^2)} = 0.008 \text{ V/m} \quad (59)$$

With both results, according to the evaluation of the Table 11 and

Table 8

Distance among circuits with a single core directly buried cable.

Number of circuits	Touching	0.2 m	0.4 m	0.6 m	0.8 m
2	0.73	0.83	0.88	0.90	0.92
3	0.60	0.73	0.79	0.83	0.86
4	0.54	0.68	0.75	0.80	0.84
5	0.49	0.63	0.72	0.78	0.82
6	0.46	0.61	0.70	0.76	0.81
7	0.43	0.58	0.68	0.75	0.80
8	0.41	0.57	0.67	0.74	0.80

Table 9

Correction factor k3, for depth of laying in underground cables.

Description	K3 for direct buried cables type 1	K3 for direct buried cables type 2	K3 for direct buried cables type 3
Depth of laying (m)	1.00	1.25	1.50
Correction factor			
Single core $\leq 185 \text{ mm}^2$	0.98	0.96	0.95
Single core $> 185 \text{ mm}^2$	0.97	0.95	0.93
Three cores	0.98	0.96	0.95

Table 10

Soil thermal resistivity in Km/W.

Cable section (mm ²)	0.7	0.8	0.9	1.0	2.0	2.5	3.0
120	1.34	1.28	1.22	1.18	0.88	0.80	0.74
150	1.35	1.28	1.23	1.18	0.88	0.80	0.74
185	1.35	1.29	1.23	1.18	0.88	0.80	0.74
240	1.36	1.29	1.23	1.18	0.88	0.80	0.74
300	1.36	1.30	1.24	1.19	0.88	0.80	0.74
400	1.37	1.30	1.24	1.19	0.88	0.80	0.74

Table 12, the current capacity is affected according to the limits of the cable distance between circuits. For instance, the distance 0.75, in the Eq. (60) and Eq. (61).

$$I_{(240\text{mm}^2)} = 203.93 \times 1.0102 = 206A \quad (60)$$

$$I_{(630\text{mm}^2)} = 313.21A \times 0.9418 = 295A \quad (61)$$

3.4. Consequence in the remaining life

The correct design of multiple cables in the same trench is of paramount importance in the field of electrical engineering, primarily to mitigate the risks associated with electric and magnetic field interactions. The significance of this aspect cannot be overstated, as it directly impacts the longevity and reliability of underground cable systems, as well as the overall safety of the electrical infrastructure.

First and foremost, when multiple cables are placed in proximity within a trench, they can create complex electromagnetic fields due to the flow of electrical current through the conductors. These fields can interact in ways that lead to increased losses, elevated temperatures, and uneven distribution of electric (E) and magnetic flux forces (B). If left unmanaged, these issues can result in accelerated aging of the cables, insulation breakdown, and ultimately, costly failures in the power distribution network. Therefore, the proper design of cable spacing, separation, and alignment within the trench is crucial to minimize these adverse effects.

Furthermore, the economic implications of cable design cannot be overlooked. Premature aging and failures necessitate costly repairs and maintenance, leading to financial burdens for utility companies and potentially causing disruptions in power supply to consumers. Additionally, inefficient cable layouts can lead to increased energy losses, which, in the long run, can have a detrimental impact on energy efficiency and sustainability goals.

Moreover, in an era where sustainable energy sources and

Table 11Cable distance between circuits for the current performance for 630mm².

Distance between phases per section	0.4	0.5	0.7	0.75	0.8	0.9	1.0
630mm ²	0.8728	0.8236	0.9217	0.9418	0.9510	0.9788	0.9801
240mm ²	0.9971	1.0011	1.0075	1.0102	1.0300	1.0482	1.0599

electrification are on the rise, optimizing cable designs becomes even more critical. Underground power transmission and distribution systems are essential for integrating renewable energy sources, such as wind and solar, into the grid. A well-designed cable system ensures efficient power delivery, minimizes losses, and enhances the overall reliability of the electrical network, thus facilitating the transition to a cleaner and more resilient energy infrastructure.

In conclusion, the correct design of multiple cables in the same trench is a foundational aspect of electrical engineering that goes beyond the technical realm; it impacts reliability, safety, economics, and sustainability. Addressing the reviewer's comment by substantiating the relevance of such studies with numerical analysis is not only essential for academic rigor but also for the continued advancement of electrical infrastructure and the promotion of a sustainable and reliable energy future for the prediction of the aging according to the configuration of the design. In Fig. 13, the blue curve is a unique circuit in a trench with laboratory conditions with a maximum life of 37 years with operation at 90 °C and 563A in a periodic condition. In the curve grey, the current is reduced in 354A with eight circuits with 1 m of separation, the maximum remaining life is 25 years, finally, the orange-blue curve has the current of 354A with a separation of 250 mm between circuits, with a remaining life of 8 years.

With the laboratory results, the baseline is indicated in the Eq. (62) regarding 1 circuit of 630 mm² in a trench with a current capacity of 563A, with the condition of degradation of the underground cable of the underground cable (D%) and the timeline in years (x). Besides, in the case of several underground cables in the same trench with a separation of 100 cm between circuits the Eq. (63) applies for Fig. 13, and the Eq. (64) represents eight circuits with a separation of 25 cm.

$$D\%_{1\text{circuit}-563A} = -0.0462x^2 - 0.8559x + 98.521 \quad (62)$$

$$D\%_{8\text{circuits}-354A-100\text{cm}} = -0.041x^2 - 2.3856x + 96.698 \quad (63)$$

$$D\%_{8\text{circuits}-354A-25\text{cm}} = -1.2317x^2 - 1.4349x + 94.657 \quad (64)$$

4. Discussion

According to the Ref. [11], in the underground line with XLPE cable, during the first minutes during the operation was found the fault, at a given isothermal temperature. And the author in the Ref. [12] showed that shield conductors are prone to penetrate the outer sheath, however, it suggests that moderate sheath penetration could also have some advantage, since it would reduce the mechanical torque caused by different temperatures throughout the cable, and the different coefficients of thermal expansion of its materials. Besides, an elastic deformation occurs when the material is deformed by stress and is then reversible, returning to its original shape when the stress is removed. If the elastic limits exceeded the capacity, inelastic deformations would occur. With increasing temperature, the unrestrained expansion of XLPE is about ten times that of Copper or Aluminum, and since XLPE is more elastic than metal, while becoming softer at higher temperatures, the

Table 12

Trench distance to reduce the effect of the induced voltage, circulating current.

Trench distance per cable section	0.5	1	2	3
630mm ²	0.6012	0.8175	0.956	1.000
240mm ²	0.7386	0.8692	1.000	1.000

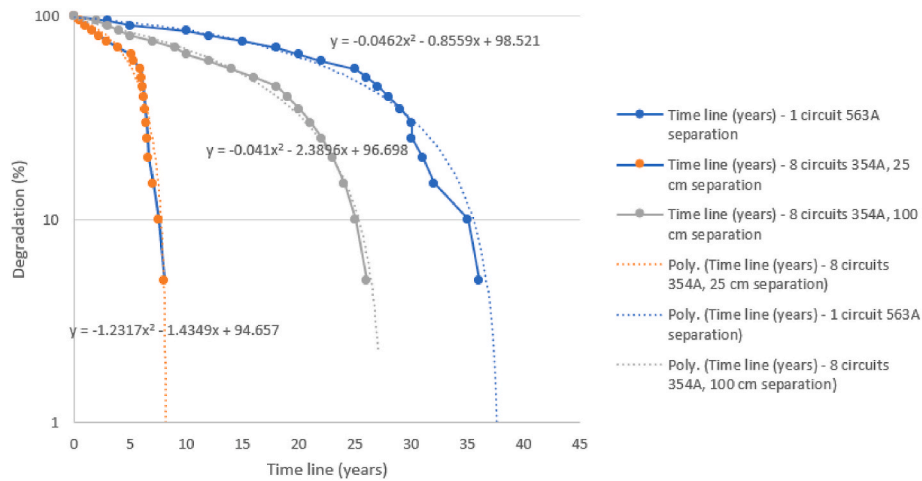


Fig. 13. The degradation and aging of the underground cable for the laboratory condition and several circuits in a trench.

insulation will then be held in position by the pressure generated. For the six case studies: Case 1 (one circuit), case 2, (two circuits), case 3 (three circuits), case 4 (four circuits), case 5 (six circuits) and case 6 (eight circuits), the soil temperature reduced its values in the Fig. 14 A), according to the depth of the underground cable, however, in the Fig. 14. B) the thermal resistance has several values in the depth of the cable. Besides, in the underground cables, the semiconductor layers are also polymer compounds, and their chemical and physical degradation is to be expected, in the same way as the insulation itself; in this sense, the mechanical effects are also important to consider. When the wire is heated, the different constituent materials do not expand equally, due to temperature differences along the length of the wire, and different coefficients of thermal expansion.

Therefore, the conductors that form the screen filaments are liable to penetrate or become dangerously embedded inside the semiconductor screens of thermoplastic material, which will lead to an increase in the intensity of the local electric field in the main insulation, affecting its durability. On the other hand, for thermo-strengthened (crosslinked) semiconductor screens, this penetration didn't consider sufficient to affect the expected lifetime, even when the cable cycled up to 145 °C for more than 200 days.

The evaluation with finite element modelling, with the information detailed in the trench has reduced the cables' current capacity though the configuration of the cables with a sudden direction of the eight circuits from 563 A to 145 A, in the Table 13.

The evaluation of the classifier algorithms to determine the capacity to 90 °C with the characteristics are tested in several techniques for the correlation evaluation and the estimation of the maximum current

Table 13

Evaluation of the cable circuits.

Description of the model	Current capacity (A)	Temperature of operation (°C)	Efficiency of the solution (%)
One circuit with external soil, with one layer	563	90	100.00
One circuit with native soil and several layers	350	90	70.00
Two circuits with native soil and several layers	268	90	47.60
Three circuits with native soil and several layers	187–252	90	33.21–44.76
Eight circuits with native soil and several layers	145–174	90	25.75–30.91

capacity in the Table 14, with the recommended algorithm with the MRFC, IBK and TREE algorithms.

The selection of the evaluation according to the tree evaluation of the algorithms 1 to 4, in the Fig. 15, with the decision-making process and the correlation (Fig. 16) of the predictive and real data.

Finally, the evaluation of eight circuits should consider the separation of 1 m between circuits to reduce the temperature of the underground cable from 191 °C (Fig. 17 A) to 90 °C (Fig. 17 B)) for the transmission of 206A; with a reduction of 100 % of the capacity to 38.4

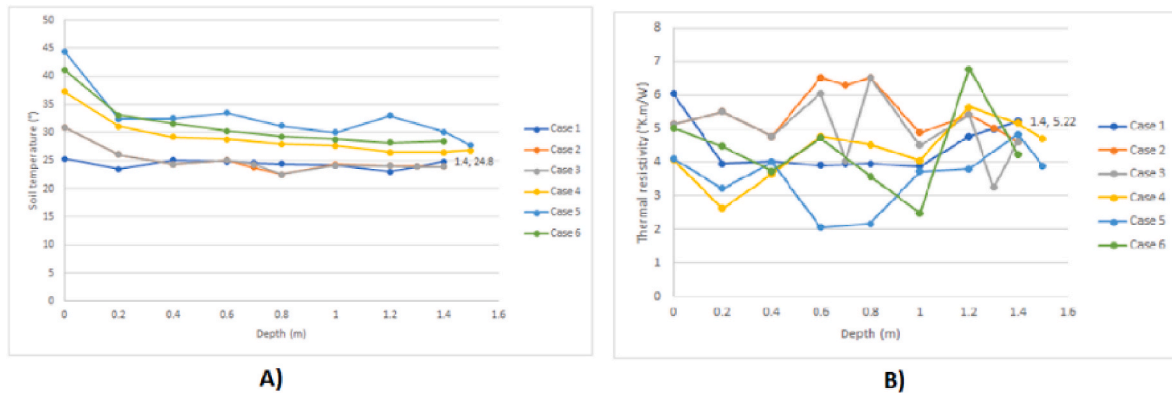


Fig. 14. A) Soil temperature according to the depth of the trench for the underground cable. B) The thermal resistance value for the case study according to the depth.

Table 14
Correlation analysis with sixteen algorithms.

Item	Algorithms	Correlation (%)	Mean absolute error (A)	Root mean squared error (A)	Relative absolute error (%)
1	Meta randomizable filtered classifier (MRFC)	97.41	6.913	22.4349	8.4723
2	Lay lBk (lBK)	97.41	6.913	22.4349	8.4723
3	Random tree (TREE)	97.40	7.17	22.4524	8.792
4	Random committee	97.40	7.3391	22.4576	8.9945
5	Lazy kStar	97.19	10.4756	23.466	12.8384
6	Additive regression	95.94	17.2147	28.0594	21.0976
7	Random forest	95.48	22.414	35.336	27.4696
8	Decision tree	93.26	29.289	36.0857	35.8954
9	Multilayer perceptron	92.06	63.0688	70.7738	77.2945
10	Lazy LWL	89.02	38.3249	46.4066	46.9694
11	Linear regression	88.81	36.2228	45.5741	44.3931
12	SMO reg	88.42	29.6268	47.5921	36.3094
13	Gaussian Process	87.70	42.9531	55.7519	52.6415
14	Random sub space	86.37	60.497	77.0104	74.1426
15	Regression by discretization	85.08	32.7822	51.1011	40.1764
16	Linear regression	52.47	62.3522	84.3963	76.4162

%.

Therefore, the spectral element approach is employed to achieve great accuracy. Interestingly, In the reference [20] certain geometrical and material factors with circular dielectric cylinders, resonance behavior described fluctuations in the parameters defining corrugated cylinders in the electric and magnetic field [20].

4.1. Scanning electron microscopy analysis for the thermal fault for pre-condition and post fault

Subsequently, a Scanning Electron Microscopy (SEM) and Energy

Dispersive X-Ray Spectroscopy (EDX) analysis were performed to evaluate the morphology before and after the fault, for this purpose, the XLPE cables were cut into 0.5 mm thick and 3 cm diameter slices of an intact cable and the cable after the fault, in the Fig. 18.

Therefore, in the Fig. 19 A-1 and Figs. 19 B-1, introduce the SEM and EDX analysis respectively of the polyethylene jacket of the cable before the fault, the results demonstrate the elemental analysis of this material shows the chemistry of C, O, Na, and Cl composition. After the event, small spots are observed in the SEM analysis (Figs. 19 A-2, Fig. 19 B-2), evaluate the copper detached from the cable, because of the fault, this is confirmed with the EDX analysis where it is observed that it is present in the Cu composition with a weight percentage (Wt%) of 1.04, which was not present before the fault.

Besides, the Fig. 20 A-1 shows the SEM analysis of the concentric copper wire of the XLPE cable, before of the fault, in this perspective, it presents amorphous, however defined characteristics, the elemental analysis by EDX (Figs. 20 A-2), it shows the composition of this cable, it is mainly Cu with Wt% of 75.05, C and O were also found in smaller proportion. After the Fault, it observed the Cu morphologically (Figs. 20 B-1), it introduces damages in all the surfaces with cracks, making the elemental analysis by EDX (Figs. 20 B-2), it shows its composition a Wt% of copper of 63.98, in addition, there is an increase in Wt% of C and O, which is because in the fault parts of the jacket and XLPE was detached, likewise, it is observed the presence in a smaller proportion of Cl and Si. At the same time, in the Fig. 21 A-1 shows the SEM analysis of the XLPE

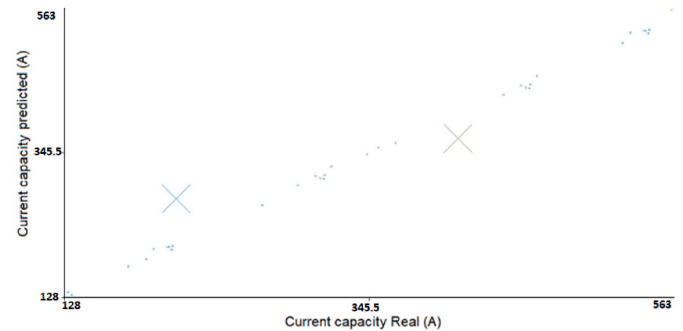


Fig. 16. Correlation of the analysis for the current capacity and the predictive current with the configuration of the circuits in the trench.

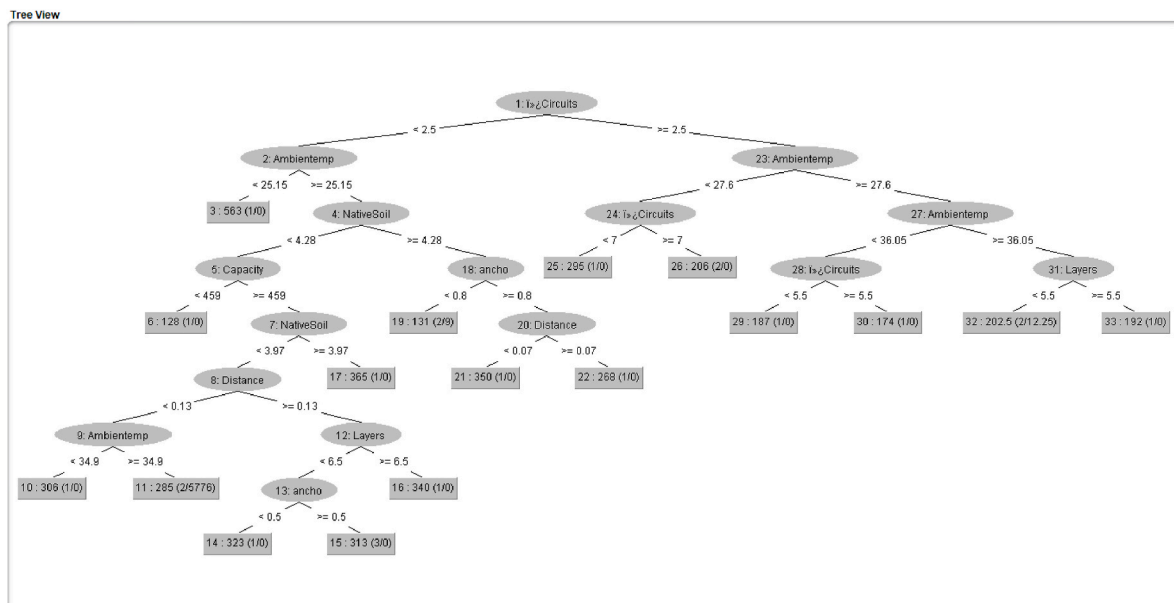


Fig. 15. Decision tree for the decision-making process in the evaluation of the correct distance and the reduction of the capacity.

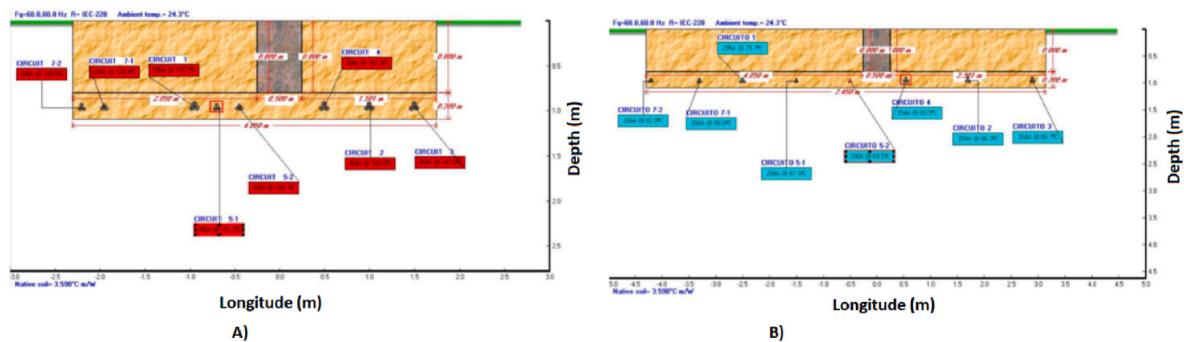


Fig. 17. Solution of the improvement according to the distance between circuits. A) Separation of 25 cm, B) Separation of 100 cm between circuits.

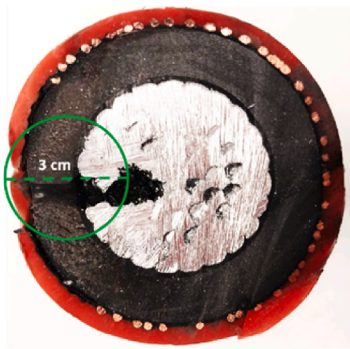


Fig. 18. Cable slice diameter for Scanning Electron Microscopy (SEM) and Energy Dispersive X-Ray Spectroscopy (EDX) analysis of the cable before and after the fault.

before the fault, in this case, the EDX analysis (Figs. 21 A-2) shows a main C and O composition with Wt% of 26.94 and 72.11. After the event, it is observed in the SEM analysis (Figs. 21 B-1) that the XLPE shows small embedded white spots which would correspond to the Cu and Al product of the fault, this was confirmed with the EDX analysis (Figs. 21 B-1), therefore, it is observed that in addition to C and O in proportions of Wt% of 16.24 and 48.70, Al and Cu are present with Wt% of 1.24 and 33.56 showing that after the event there is significant damage in the cable. Finally, in the Fig. 22 A-1 shows the SEM analysis of the aluminum conductor before the fault, which creates an elemental composition determined by EDX of C, O, and Al with Wt% of 4.97, 51.74, and 43.29. After the fault around the aluminum conductor, two regions are distinguished, being Region 1 the area most affected by the event which is observed in the SEM analysis in the Figs. 22 B-1, in this same figure Region 2 is observed where the aluminum conductor is observed with volume losses and deformations after the morphological analysis. The EDX analysis shows that Region 1 (Figs. 22 B-2) presents in its composition C and O mainly with Wt% of 22.63 and 68.13 respectively which would correspond to the XLPE melted after the cable fault.

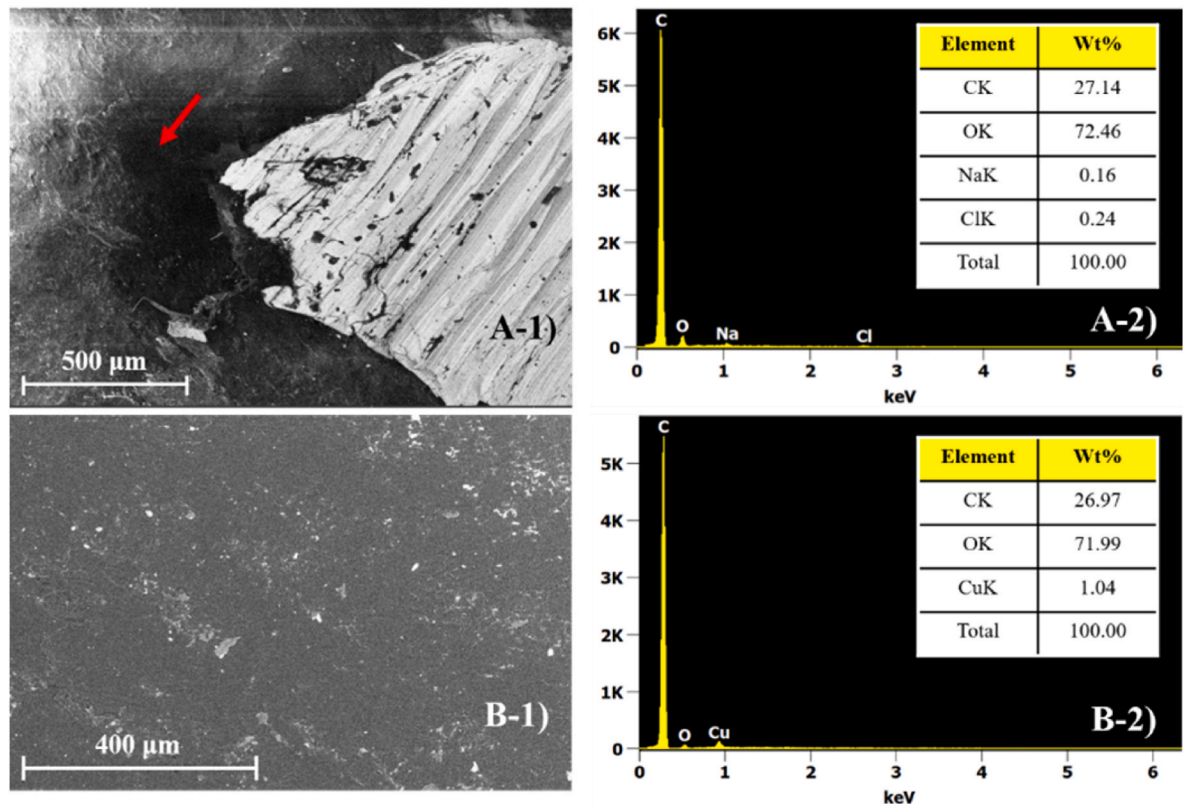


Fig. 19. SEM image of the polyethylene jacket before (A-1) and after the event (B-1). EDX of the polyethylene jacket before (A-2) and after the event (B-2).

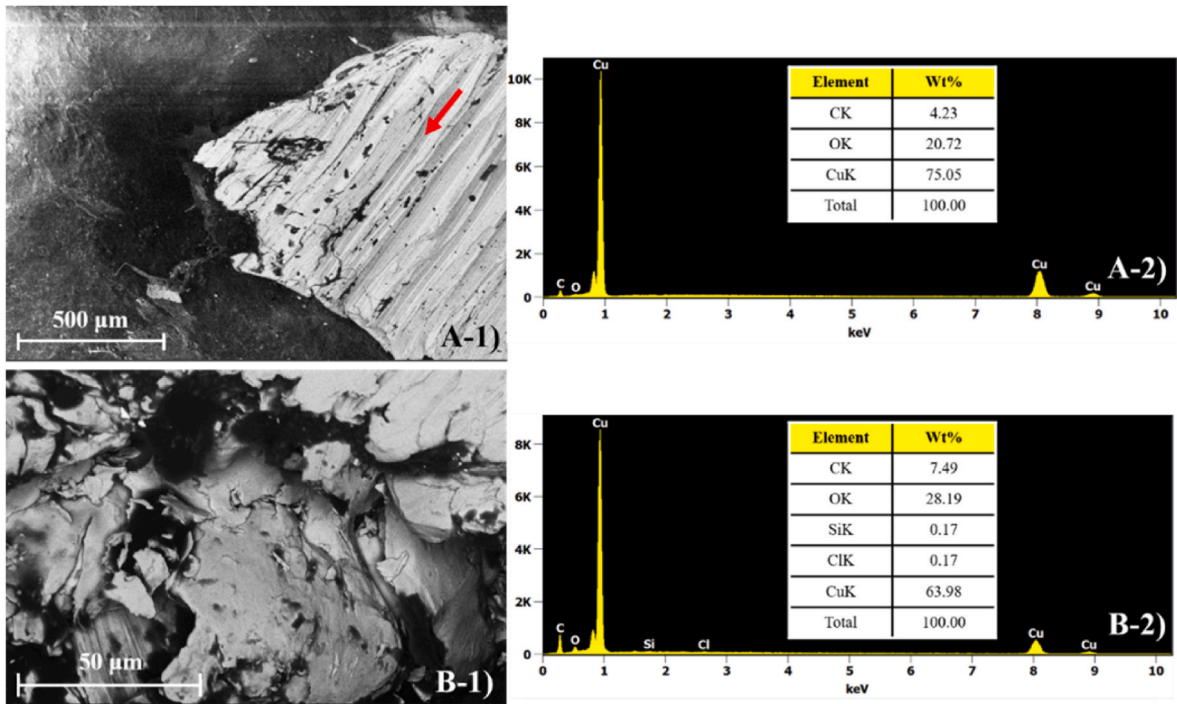


Fig. 20. SEM image of concentric copper wire before (A-1) and after the fault (B-1). EDX of concentric copper wire before (A-2) and after the fault (B-2).

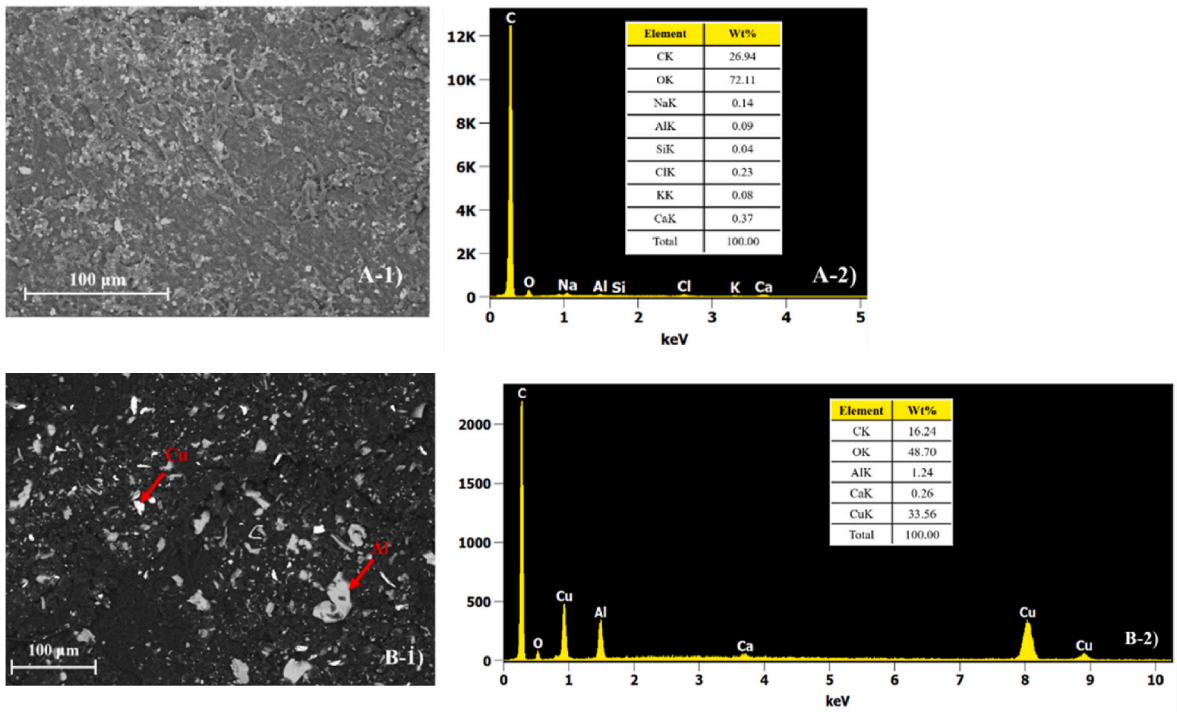


Fig. 21. SEM image of XLPE before (A-1) and after the fault (B-1). EDX of the XLPE before (A-2) and after the fault (B-2).

In addition, in Fig. 20 A-1 and B-1, there is evidence of Al with Wt% of 8.49 that corresponds to the aluminum conductor detached in the fault. An EDX analysis of Region 2 also shows that the elemental composition of the aluminum conductor is Al, O, and C with Wt% of 42.10, 53.32, and 5.58 respectively, in Fig. 21.

4.2. Recommendation for the improvement in the distance between circuits

Fig. 23, the same configuration for five circuits in a trench for the section of 630mm2 allows to obtain 295A in the simulation. However, if some additional circuits are required, the reduction of the current capacity drops dramatically.

The evaluation in the Fig. 24, incorporates the five circuits in the first

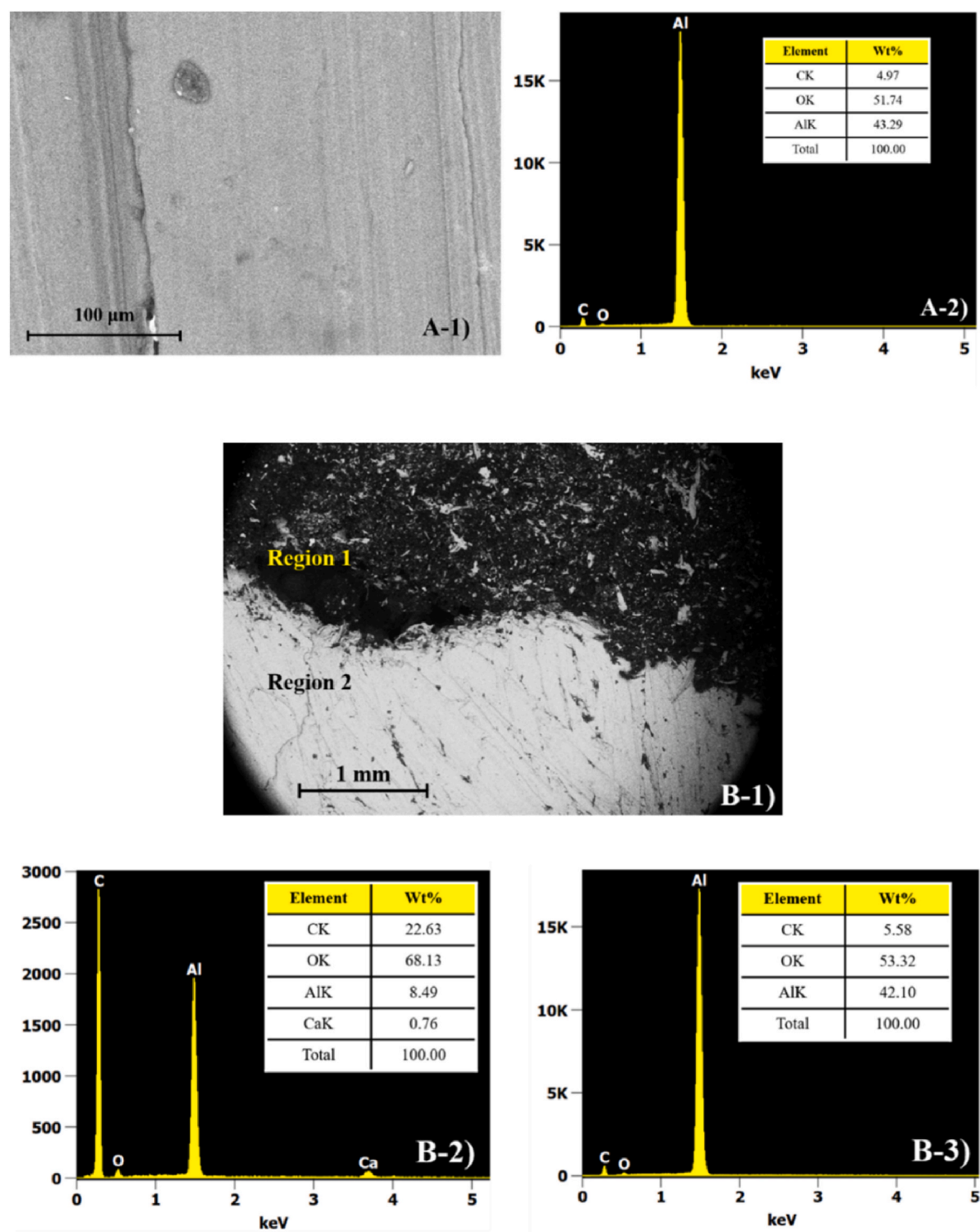


Fig. 22. SEM image of the aluminum conductor before (A-1) and after the fault showing two regions (B-1). EDX of the aluminum conductor before (A-2) and after the fault of region 1 (B-2), and region 2 (B-3).

trench of the right side with a section of 240mm², and the others three are in the left side separated 2 m according to the.

Therefore, similar the magnetic field connected to the electrical overhead lines links a steel tower and induces eddy currents; depending on the electrical permittivity, magnetic permeability, and carrying current, this eddy current may produce significant losses in the tower [21], though the loss is minimal at equilibrium, in underground cables the insulation, distance to the circuits and ground thermal resistivity have influence in the maximum capacity and the maximum temperature

of the cables. Lastly, and perhaps most crucially, this study offers a thorough understanding of how to minimize related losses by generating electric and magnetic field depending on the underground circuit design.

4.3. Comparison of the results for electrodynamic modeling and the technical characteristics

The comparison of the results with a specialized software Electrical

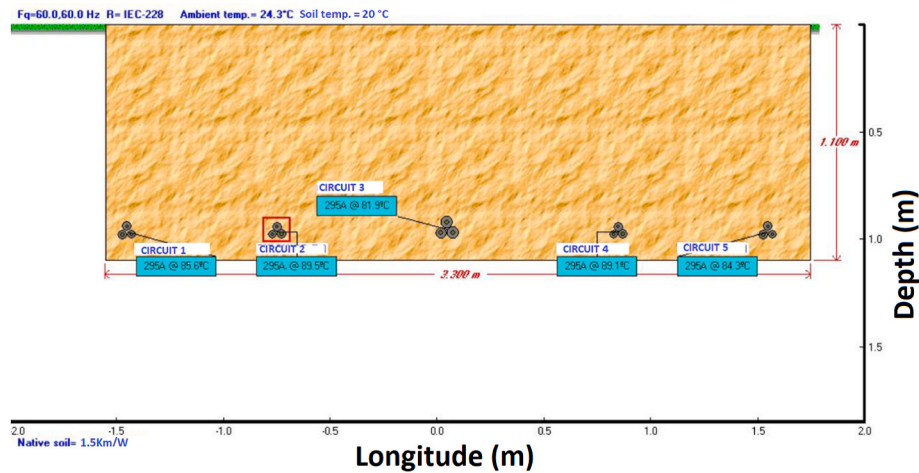


Fig. 23. Evaluation with finite elements for the condition calculated associated with 5 circuits of 630mm².

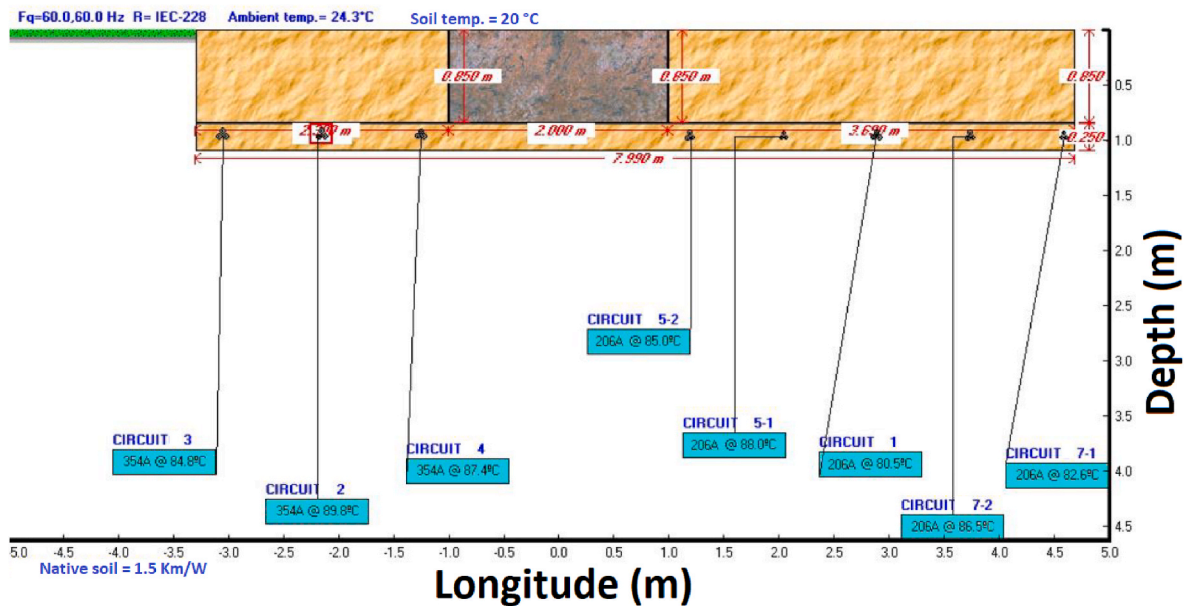


Fig. 24. Evaluation of the separation for the trench in 5 circuits of 240 mm² and a second trench with 630 mm².

Transient Analyzer Program (ETAP), with the method Neher-McGrath, the difference between the finite element modelling compared with the electrodynamic modeling with the characteristics indicated in the Table 16, the baseline of one circuit in a trench with one layer, it has a difference of 2.31 % at the 90 °C, according to the Table 15.

Besides, the condition of several layers with 1 circuit, in both methods, the difference is 2.00 % and with two circuits with several layers, the difference is 2.61 %. However, out of the scope of the international standard, the difference between the finite element and the ETAP process is 6.42 %–7.54 % for three circuits and 6.90 %–7.47 % for eight circuits.

The simulations has the following recommendations: An improvement in the insulation thickness (rated/minimum) with comparison to the traditional ones of 8mm², in Table 16. The cost with the manufacturer is higher only by 1.018 % with the local market. In this case, the quantity and separation of the cables with this improvement in the insulation, allows to increase the remaining life to 8 years–26 years for eight circuits in the same trench by separating the circuits from 25 cm to 100 cm, according to Fig. 13. The operating and abnormal modes of the system determine the range of thickness for the XLPE insulation material

surrounding the covered conductor cable in the medium voltage transmission lines. Regarding earlier research, the XLPE thickness (d) is set at 3 mm since this value yields the best results when calculating electric fields and the economics of covered conductors throughout the whole range of (d). Additionally, the distance (D) between adjacent covered conductors in free space, which is crucial for the economic analysis of XLPE, is set in this study at a value of 1 m since it produces the best results out of all the possible values in the (D) range [17].

5. Conclusion

The main conclusions are significant challenges associated with the design of medium voltage underground circuits, particularly in the context of renewable energy projects. The study also established the accuracy of the proposed research process, achieving a high level of precision, with results showing a precision of 99.4850 % $\pm$ 0.5150 % for 240 mm² and 99.3698 % $\pm$ 0.6301 % for 630 mm². These conclusions emphasize the importance of addressing the limitations in current international standards and considering factors like soil conditions when designing medium voltage underground circuits for renewable energy

Table 15
Evaluation of the cable circuits with finite elements, and ETAP.

Description of the model	Current capacity – Finite element (A)	Temperature of operation (°C)	ETAP Current capacity (A)	Difference in absolute values (%)
One circuit with external soil, with one layer	563	90	550	2.3090
One circuit with native soil and several layers	350	90	357	2.0000
Two circuits with native soil and several layers	268	90	275	2.6119
Three circuits with native soil and several layers	187–252	90	199–271	(6.4171–7.5396)
Eight circuits with native soil and several layers	145–174	90	155–187	(6.8965–7.4712)

Table 16
Recommended Technical characteristics.

Description	Unit	240mm2	630mm2
Phase/phase voltage	kV	33	33
Rated voltage	kV	20/34.5	20/34.5
Maximum voltage	kV	37.95	37.95
Rated short-term withstand voltage at power frequency	kV	70	70
Withstand voltage assigned to lightning impulse	kVp	170	170
Standard	–	IEC 60502-2	IEC 60502-2
Material	–	AAC 1350 (Al)	AAC 1350 (Al)
Nominal section	mm ²	240	630
Minimum strand count	–	30	53
Diameter (minimum – maximum)	mm	17.6–19.2	28.7/32.5
Cable resistance (20 °C)	Ω/km	0.125	0.0469
Shield cable thickness (rated/ minimum)	mm	≥0.5/0.3	≥0.5/0.3
Insulation material	–	XLPE	XLPE
Insulation thickness (rated/ minimum)	mm	8.8/7.82	8.8/7.82
Insulation screen thickness (rated/ minimum)	mm	≥0.5/0.3	≥0.5/0.3
Longitudinal block against water minimal overlap	%	10	10
Ground shield (aluminum belt) thickness/diameter	mm	0.354/40.02	0.354/48.83
External insulation material	–	ST7 polyethylene	ST7 polyethylene
External insulation thickness (rated/ minimum)	mm	3/≥ 1.4	3/≥ 1.4

projects. The research provides valuable insights into these challenges and highlights the need for improvements in current standards and design practices.

The thermal influence of the combination of XLPE circuits with different distance and depth reduces the maximum current capacity with an overloading. Therefore, the distance of 15 cm between circuits reduces the capacity in eight circuits from 100 % to 25.75 %–30.91 %;

even the native soil affects the current capacity by reducing the current capacity in 30 %.

The effects of the operation with more than two XLPE circuits is a reduction of the current capacity with the effect of the thermal resistance (soil), the layers associated in the trench and the depth. The main effect of the failure mode is the high temperature in the underground cable, therefore, the radiographic indicated a venting channel skewed to one side epicenter of the fault, it is located exactly on the helical projection of the copper tape of the screen (currently volatilized), arranged for the retention of the filaments, and an increase in the conductive section with a clean perforation; due to the effect of temperature and the consequent Radial expansion of the XLPE has undermined the inner side of the outer shell, confirming the influence of the copper tape in the fault sector; it is associated a change and turbidity in its coloration associated to the shield and the insulation. Finally, the partial discharge test found voltage levels to 1.7U₀, in the internal semiconductor of the MV cable. As a summary, the partial discharge is 200 with a maximum value of 3800 pC.

Mathematical modelling improves the equations from IEC 60287, 60502, and 60228. In these standards, the quantity of underground cables is two in the same trench. However, with the proposed methodology, the quantity could increase according to the necessary quantity of cables. The case study has considered five circuits in the same trench and the evaluation of a parallel trench of three circuits, the validation of the instrument was executed with a finite element modelling; it helps in analyzing and optimizing cable performance, assessing thermal characteristics, and ensuring overall system reliability in the methodology proposed. By simulating various scenarios, it could identify potential issues in the quantity of the cables in the same trench and the distance between parallel circuits in different trenches as heat distribution, voltage drop and mechanical stress, allowing for better cable design and placement to enhance efficiency and minimize failures.

For future works, the application of the more insulation and a higher section of the shield should be considered and independent trenches for more than two circuits of XLPE, for more circuits, it should be updated in the IEC standards with the recommendations of the research to avoid the sudden failure for early degradation of the insulation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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